Chapter 6

Recovery of the Dirac Equation

Chapter **??** showed that the free Galilean Hamiltonian is not an input of the theory but a consequence of covariance. Once the Galilei group is required to act on the projective state space by Kähler isomorphisms, the relation between boosts and time translations, together with irreducibility, forces $H = |\mathbf{P}|^2/2m + E_0\mathbb{1}$ (Theorem **??**). Three features of that result will serve as a guide in the present chapter. The mass m was not put in by hand but read off from a phase: it is the parameter of a central extension, because $H^2(\mathfrak{g}, \mathbb{R}) \cong \mathbb{R}$ for the Galilei algebra (Proposition **??**). The constant E_0 remained undetermined, because a Galilean action lifts to a unitary representation only up to a character $e^{i\epsilon s}$ (Proposition **??**). And spin played no dynamical role: in the Schrödinger realization of Corollary **??** the boosts multiply the orbital factor by $e^{im\mathbf{v}\cdot\mathbf{x}/\hbar}$ and never touch the factor $\mathbb{C}^{2j+1}$, which is acted upon by rotations only.

The setting of the chapter is the one of Chapter **??**. The pure states of the system are the points of the projective Hilbert space $\mathbb{P}\mathcal{H}$ with its Kähler structure, and $\mathcal{H}$ is a separable Hilbert space which is fixed up to unitary equivalence and carries no further structure. No momentum, no position and no spin is assumed. The only additional input is a group of symmetries acting on $\mathbb{P}\mathcal{H}$ by Kähler isomorphisms, together with irreducibility and positivity of the energy. Everything else is derived from the action, and in particular the spin is an output of the construction and not a datum. The realization of $\mathcal{H}$ as a space of functions of the momentum with values in a spin space is built along the way, because the Kähler manifold alone cannot contain it, as Remark 6.3.4 shows.

The Galilei group is the symmetry group of a spacetime in which all inertial observers share the same time coordinate. This is an excellent approximation at low velocities, but it is not how nature is organized: in special relativity the time coordinate of an observer is mixed with the space coordinates by a change of inertial frame, and the group of symmetries is the Poincaré group. The strategy of Chapter **??** carries over unchanged, since it only asks that the symmetries of spacetime be implemented on the state space. What changes are the structures met along the way, and it is useful to list them in advance, because they organize the whole chapter.

First, the second cohomology of the Lie algebra vanishes. No phase is forced upon us, no central extension is needed, the lift of the action to a unitary representation is unique and the energy is fixed without additive ambiguity. We give a complete proof of the vanishing, instead of quoting it, because it explains why the Galilean mass cannot survive in this form. Second, the mass is no longer the parameter of a cocycle but the value of a Casimir operator, and the Hamiltonian $H = \sqrt{|\mathbf{P}|^2 + m^2}$ follows in a few lines (Theorem 6.3). Third, and this is the real subject of the chapter, boosts now act on the spin. The unitary representation is induced from the rotations, and a boost rotates the internal degrees of freedom by a matrix that depends on the momentum, the Wigner rotation. The representation obtained is correct and irreducible, but it is not local: the transformation of the value of a state at the momentum p is not given by a fixed matrix independent of p, and this has no Galilean counterpart.

The way out, which is also the way to the Dirac equation, is the following. We look for a finite-dimensional complex vector space W carrying a linear representation D of the Lorentz group, necessarily non-unitary, and we realize the induced representation on W-valued functions of the momentum, constrained to a subspace $F(p) \subseteq W$ that depends on p. The constraint is obtained by transporting from the rest frame, where it must be invariant under rotations. The requirement of invariance under space inversion forces W to be four-dimensional and fixes the rest-frame constraint up to a sign. The transported

constraint is the Dirac equation. As in Chapter **??**, the equation of motion is a consequence of covariance and not a postulate: the gamma matrices, the four components and the first-order form are outputs of the construction. This reverses the logic of Dirac's original argument [**TODO-Dirac1928**], where a first-order equation with a positive density was postulated.

It may help to think of the chapter as a staircase with four steps. The first step identifies the group and proves that its action can be implemented without enlarging it, which is the content of Sections 6.1 and 6.2. The second step determines the energy from the Casimir operator and the positivity of the energy, and it is carried out in Section 6.3. The third step constructs the Hilbert space above the mass shell. The spin emerges there as the label of an irreducible representation of the stabilizer of the rest momentum, and boosts act on it through the Wigner rotation, as shown in Section 6.4. The fourth step looks for a description by fields which transform pointwise. It is possible at the price of redundant components and of an equation selecting the physical ones, and this equation is Dirac's, see Sections 6.5 to 6.7.

Sections 6.1 to 6.3 form a relativistic counterpart of Chapter **??** and can be read independently of the rest. Section 6.1 introduces the Poincaré group in a matrix model, Section 6.2 proves that it has no central extensions, and Section 6.3 states the setting, lifts the action and identifies mass and energy. Section 6.4 studies the mass shell and proves that every elementary system is an induced representation labelled by the mass and the spin. The remaining sections derive the equation. We restrict to four spacetime dimensions and to $m > 0$. The massless case, higher spins and interactions are not treated.

6.1 The Poincaré Group

Before turning to representations we need to know precisely which group we are representing. In Chapter **??** the answer was the Galilei group of Section **??**, built from four kinds of transformations: translations in time, translations in space, rotations and boosts. The Poincaré group, which is the group of isometries of Minkowski spacetime, has the same four ingredients, and the difference lies entirely in the way they compose. In Newtonian spacetime a boost leaves the time coordinate alone and adds a displacement proportional to time. In Minkowski spacetime a boost is a hyperbolic rotation in the plane spanned by the time axis and the direction of motion: it mixes time and space, in the same way as an ordinary rotation mixes two space directions, but with hyperbolic functions in place of trigonometric ones. This single change is responsible for everything that distinguishes the two groups, from the Thomas precession to the vanishing of the Galilean cocycle.

This section introduces the group in the form required by the rest of the chapter. We make three choices. The first is to work with the universal covering, as we did for the Galilei group. The topology of the Lorentz group, like that of $SO(3)$, is not simply connected, and a representation on rays can be a genuine representation only of the covering, which reintroduces the sign ambiguity of Example **??**. The second choice is a concrete model. The universal covering of the proper orthochronous Lorentz group is $SL(2,\mathbb{C})$, and the covering map has an elementary realization through the action of 2×2 matrices on Hermitian matrices. This is the same mechanism by which $SU(2)$ rotates the traceless Hermitian matrices $\mathbf{x}\cdot\boldsymbol{\sigma}$, which model $\mathbb{R}^3$ in the description of a spin $\frac{1}{2}$ (Example 1.2.5 and the Bloch sphere of Section 2.7). Adding the identity matrix to the picture adds the time direction and upgrades rotations to Lorentz transformations. The third choice is to use coordinates adapted to the physics, writing every element as a translation, a boost and a rotation, and to record the relations between these transformations in the same form as Lemma **??**.

We write $\mathbb{R}^{1,3} = \mathbb{R}^4$ for Minkowski space, with points $x = (x^0, \mathbf{x})$ and bilinear form $x \cdot y := x^0 y^0 - \mathbf{x}\cdot\mathbf{y}$, which we also write $\eta(x,y)$ when we want to stress the matrix $\eta = \mathrm{diag}(1,-1,-1,-1)$. We put $x^2 := x \cdot x$. The speed of light is set equal to one until Section 6.3.4. The Lorentz group $O(1,3)$ is the Lie group of linear maps of $\mathbb{R}^{1,3}$ preserving the form, and $SO^{\uparrow}(1,3)$ denotes its connected component of the identity, which is the group of proper orthochronous Lorentz transformations, *i.e.*, those that preserve orientation and the direction of time [**TODO-BarutRaczka**]. Since $O(1,3)$ is a closed subgroup of $GL(4,\mathbb{R})$, so is $SO^{\uparrow}(1,3)$, because connected components are closed.

A Matrix Model of Minkowski Space

The model rests on the observation that a Minkowski vector can be traded for a Hermitian matrix, in such a way that the Minkowski square becomes a determinant. Determinants are preserved by matrices of determinant one, which is the origin of the covering.

Definition 6.1.1 (Hermitian matrix model of $\mathbb{R}^{1,3}$)**.** *For $x \in \mathbb{R}^{1,3}$ we set*

$$\hat{x} := x^0\,\mathbb{1} + \mathbf{x}\cdot\boldsymbol{\sigma}, \qquad \tilde{x} := x^0\,\mathbb{1} - \mathbf{x}\cdot\boldsymbol{\sigma}, \qquad \mathbf{x}\cdot\boldsymbol{\sigma} = \sum_{j=1}^{3} x^j\sigma_j, \tag{6.1}$$

where σ_j are the Pauli matrices of Example 1.2.5.

The matrix $\hat{x}$ is the model proper. The matrix $\tilde{x}$ is the same vector with the spatial part reversed, *i.e.*, the image of x under space inversion, and it will be indispensable in Section 6.5, where space inversion exchanges two inequivalent spinor representations.

Lemma 6.1.2 (Algebra of the matrix model)**.** *1. The map $x \mapsto \hat{x}$ is a real linear bijection from $\mathbb{R}^{1,3}$ onto the real vector space of Hermitian 2×2 matrices, and $\det\hat{x} = x^2$.*

2. $\tilde{x} = \operatorname{adj}\hat{x}$, $\hat{x}\tilde{x} = \tilde{x}\hat{x} = x^2\,\mathbb{1}$ and $x\cdot y = \frac{1}{2}\operatorname{tr}(\hat{x}\tilde{y})$.

3. If $x^2 > 0$, then $\hat{x}$ is positive definite if $x^0 > 0$ and negative definite if $x^0 < 0$.

Proof. Item 1. A Hermitian 2×2 matrix has the form $\begin{pmatrix} x^0+x^3 & x^1-ix^2 \\ x^1+ix^2 & x^0-x^3 \end{pmatrix}$ with real x^μ, and its determinant is $(x^0)^2 - |\mathbf{x}|^2$. Item 2. For a 2×2 matrix the adjugate is $\operatorname{adj}X = (\operatorname{tr}X)\mathbb{1} - X$, so that $\operatorname{adj}\hat{x} = \tilde{x}$, and $X\operatorname{adj}X = \det X\,\mathbb{1}$. The last identity follows from $(\mathbf{x}\cdot\boldsymbol{\sigma})(\mathbf{y}\cdot\boldsymbol{\sigma}) = \mathbf{x}\cdot\mathbf{y}\,\mathbb{1} + i(\mathbf{x}\times\mathbf{y})\cdot\boldsymbol{\sigma}$. Item 3. The eigenvalues of $\hat{x}$ are $x^0 \pm |\mathbf{x}|$. Their product is $x^2 > 0$, so they have the same sign, which is the sign of their sum $2x^0$. □

Boosts and Rotations: the Polar Decomposition

The covering map is built from the action $\hat{x} \mapsto g\hat{x}g^\dagger$, and to prove that it covers the whole proper orthochronous Lorentz group we need to know that $SL(2,\mathbb{C})$ is connected. We therefore begin with the polar decomposition of its elements. Besides being the technical tool, it has a clear physical meaning, which we state right after the proof.

Lemma 6.1.3 (Polar decomposition)**.** *The map $\Psi : \mathbb{R}^3 \times SU(2) \to SL(2,\mathbb{C})$ defined by*

$$\Psi(\mathbf{w}, u) := \exp\!\big(\tfrac{1}{2}\,\mathbf{w}\cdot\boldsymbol{\sigma}\big)\,u \tag{6.2}$$

is a diffeomorphism. In particular $SL(2,\mathbb{C})$ is connected and simply connected.

Proof. Let $g \in SL(2,\mathbb{C})$. The matrix $gg^\dagger$ is Hermitian and positive definite, with determinant one. It has a unique positive definite Hermitian square root $H := (gg^\dagger)^{1/2}$, which depends smoothly on g and has $\det H = 1$, because the eigenvalues of H are the positive square roots of those of $gg^\dagger$. The matrix $u := H^{-1}g$ satisfies $uu^\dagger = H^{-1}gg^\dagger H^{-1} = \mathbb{1}$ and $\det u = 1$, so that $u \in SU(2)$ and $g = Hu$. If $g = H'u'$ is another decomposition of this type, then $gg^\dagger = H'^2$, and the uniqueness of the positive square root yields $H' = H$ and $u' = u$.

It remains to parametrize the matrices H. A positive definite Hermitian matrix is the exponential of a unique Hermitian matrix $h = \log H$, which depends smoothly on H, and $\operatorname{tr}h = \log\det H = 0$ because $\det e^h = e^{\operatorname{tr}h}$. Traceless Hermitian matrices are exactly the matrices $\frac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}$ with a unique $\mathbf{w} \in \mathbb{R}^3$, which shows that Ψ is a smooth bijection with smooth inverse. The group $SU(2)$ is diffeomorphic to S^3, since it consists of the matrices with columns $(\alpha,\beta)^T$ and $(-\bar{\beta},\bar{\alpha})^T$ subject to $|\alpha|^2 + |\beta|^2 = 1$. Hence $SL(2,\mathbb{C}) \simeq \mathbb{R}^3 \times S^3$ is connected and simply connected. □

See [**TODO-Hal03**] and [**TODO-BarutRaczka**] for the polar and Cartan decompositions in general.

Remark 6.1.4 (Every Lorentz transformation is a boost followed by a rotation). The lemma says that every element of $SL(2,\mathbb{C})$ is uniquely a Hermitian positive factor $\exp\left(\frac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\right)$ times a unitary one. Anticipating Lemma 6.1.5, the second factor is a rotation and the first is a pure boost, with rapidity vector $\mathbf{w}$. Thus every proper orthochronous Lorentz transformation is a pure boost composed with a rotation, in a unique way, and the topology of the group is the product of the topology of the boosts, a $\mathbb{R}^3$, and that of the rotations, an S^3. This is the same decomposition that organizes the Galilei group, whose underlying manifold is $\mathbb{R}^9 \times S^3$ by Definition **??**. The boosts are the non-compact part of the group, and all the novelties of the Lorentz group with respect to the rotation group of the previous chapters come from there.

The Covering of the Lorentz Group

With the connectedness of $SL(2,\mathbb{C})$ at hand we can construct the covering map. The following lemma is the relativistic analogue of the covering $SU(2) \to SO(3)$ that entered the definition of the Galilei group, and it also yields the differential of the map, which is needed to compute the Lie algebra, and the covariance of the matrices $\tilde{x}$, which is needed in Section 6.5.

Lemma 6.1.5 (Covering of the Lorentz group). *For $g \in SL(2,\mathbb{C})$ let $\Lambda(g) : \mathbb{R}^{1,3} \to \mathbb{R}^{1,3}$ be the linear map defined by*

$$\widehat{\Lambda(g)x} = g\,\hat{x}\,g^\dagger. \tag{6.3}$$

1. *$g \mapsto \Lambda(g)$ is a smooth surjective homomorphism $SL(2,\mathbb{C}) \to SO^\uparrow(1,3)$ with kernel $\{\mathbb{1}, -\mathbb{1}\}$, and hence a covering map.*

2. *Its differential at the identity is the injective linear map $\rho : \mathfrak{sl}(2,\mathbb{C}) \to \mathfrak{gl}(\mathbb{R}^{1,3})$ defined by*

$$\widehat{\rho(\xi)x} = \xi\,\hat{x} + \hat{x}\,\xi^\dagger, \tag{6.4}$$

   where $\mathfrak{sl}(2,\mathbb{C})$ is regarded as a real Lie algebra of dimension 6.

3. *For all g and x,*

$$\widetilde{\Lambda(g)x} = (g^\dagger)^{-1}\,\tilde{x}\,g^{-1}. \tag{6.5}$$

Proof. Item 1. The matrix $g\hat{x}g^\dagger$ is Hermitian, so Equation (6.3) defines a linear map $\Lambda(g)$. It preserves $x^2 = \det\hat{x}$ because $\det\left(g\hat{x}g^\dagger\right) = |\det g|^2 \det\hat{x} = \det\hat{x}$, and it preserves the bilinear form by polarization, so $\Lambda(g) \in O(1,3)$. The relation $\Lambda(gg') = \Lambda(g)\Lambda(g')$ is immediate and $g \mapsto \Lambda(g)$ is polynomial in the entries of g. By Lemma 6.1.3 the group $SL(2,\mathbb{C})$ is connected, hence its image lies in the connected component $SO^\uparrow(1,3)$.

Item 2. Differentiating $g(t)\hat{x}g(t)^\dagger$ along $g(t) = \exp(t\xi)$ at $t = 0$ yields Equation (6.4). If $\rho(\xi) = 0$, then $x = (1,\mathbf{0})$ yields $\xi + \xi^\dagger = 0$, and $x = (0,\mathbf{e}_j)$ then yields $\xi\sigma_j - \sigma_j\xi = 0$ for $j = 1,2,3$. The matrix ξ commutes with $\mathbb{1}, \sigma_1, \sigma_2, \sigma_3$, which span $M_2(\mathbb{C})$, so ξ is a multiple of the identity, and $\operatorname{tr}\xi = 0$ forces $\xi = 0$. Hence ρ is injective. The Lie algebra of $O(1,3)$ is $\{M \in M_4(\mathbb{R}) \mid \eta M + M^T\eta = 0\}$, where ηM is antisymmetric, so it has dimension $6 = \dim\mathfrak{sl}(2,\mathbb{C})$. The image of ρ lies in it, since $\Lambda(\exp t\xi) = \exp(t\rho(\xi))$ is in $O(1,3)$. It descends that ρ is a linear bijection onto this Lie algebra, and by the inverse function theorem $g \mapsto \Lambda(g)$ maps a neighbourhood of $\mathbb{1}$ diffeomorphically onto a neighbourhood of the identity of $SO^\uparrow(1,3)$. The image is a subgroup containing this neighbourhood, and a neighbourhood of the identity in a connected group generates the whole group, so $g \mapsto \Lambda(g)$ is surjective.

If $\Lambda(g) = \mathrm{id}$, then $gg^\dagger = \mathbb{1}$ from $x = (1,\mathbf{0})$ and $g\sigma_j g^{-1} = \sigma_j$ for all j, so that g commutes with $M_2(\mathbb{C})$ and $g = \lambda\mathbb{1}$ with $\lambda^2 = \det g = 1$. The kernel is therefore $\{\mathbb{1}, -\mathbb{1}\}$. A surjective Lie group homomorphism with discrete kernel is a covering map [**TODO-Lee13**].

Item 3. By item 2 of Lemma 6.1.2, $\operatorname{adj}(XY) = \operatorname{adj}(Y)\operatorname{adj}(X)$ and $\operatorname{adj} g = g^{-1}$ for $\det g = 1$, so $\widetilde{\Lambda(g)x} = \operatorname{adj}(g\hat{x}g^\dagger) = \operatorname{adj}(g^\dagger)\,\tilde{x}\,\operatorname{adj}(g) = (g^\dagger)^{-1}\tilde{x}\,g^{-1}$. □

This is the covering used in the definition of the Poincaré group, in the same way as $SU(2) \to SO(3)$ is used in Definition **??**. Restricted to $SU(2)$ it is that map, since $u\hat{x}u^\dagger = x^0\mathbb{1} + (R_u\mathbf{x})\cdot\boldsymbol{\sigma}$ for $u \in SU(2)$.

Remark 6.1.6 (Why the covering, and what it has to do with spin). The kernel $\{\pm\mathbb{1}\}$ is the matrix counterpart of the fact that a rotation by 2π is the identity on vectors but is $-\mathbb{1}$ on spinors, as in Example **??**. The Lorentz group has the same fundamental group as the rotation group, $\mathbb{Z}_2$, because the boosts are contractible and only the rotations contribute (Remark 6.1.4). Thus nothing new happens topologically when passing from rotations to Lorentz transformations: the sign ambiguity is the same, and it is removed in the same way, by going to the covering. We will see in Proposition 6.2.5 and Corollary 6.4.10 that the value taken by the central element $-\mathbb{1}$ in a representation is precisely what distinguishes integer from half-integer spin.

The Group and its Elementary Transformations

The covering map allows us to define the group of isometries of Minkowski space together with its universal covering. We follow the construction of Definition **??**, in which the rotation group acts on the translation parameters and the product is a semidirect one.

Definition 6.1.7 (Poincaré group)**.** *Let $\Lambda(g)$ be as per Lemma 6.1.5. The* Poincaré group *$\mathcal{P}$ is the Lie group with underlying manifold $\mathbb{R}^{1,3} \times SL(2,\mathbb{C})$ and product*

$$(a,g)(a',g') := \big(a + \Lambda(g)a',\ gg'\big). \tag{6.6}$$

The product is the one of the semidirect product $\mathbb{R}^{1,3} \rtimes SL(2,\mathbb{C})$ for the action $g \mapsto \Lambda(g)$ by automorphisms of the additive group of $\mathbb{R}^{1,3}$ [**TODO-BarutRaczka**]. Translations form a normal subgroup, which is abelian, and the Lorentz part acts on it by conjugation. The same pattern holds for the Galilei group, whose structure is that of Definition **??**. The next remark verifies that the definition yields a Lie group and relates it to the geometry of Minkowski space.

Remark 6.1.8. Bracketing a triple product in the two possible ways yields in both cases $(a + \Lambda(g)a' + \Lambda(gg')a'',\ gg'g'')$, since $\Lambda(gg') = \Lambda(g)\Lambda(g')$, so the product is associative. The identity is $(0,\mathbb{1})$ and the inverse of (a,g) is $(-\Lambda(g^{-1})a,\ g^{-1})$. These operations are smooth because they are polynomial in the entries of g and in a, and $\mathcal{P}$ is therefore a Lie group of dimension 10. The group acts on $\mathbb{R}^{1,3}$ by $(a,g)\cdot x := \Lambda(g)x + a$, and this is an action in the sense of Definition 1.5.1, because $\big((a,g)(a',g')\big)\cdot x = \Lambda(gg')x + a + \Lambda(g)a' = (a,g)\cdot\big((a',g')\cdot x\big)$. The kernel of the action is $\{(0,\mathbb{1}),(0,-\mathbb{1})\}$, so that the quotient $\bar{\mathcal{P}}_0 := \mathcal{P}/\{(0,\pm\mathbb{1})\} \simeq \mathbb{R}^{1,3} \rtimes SO^{\uparrow}(1,3)$ is the group of proper orthochronous isometries of Minkowski space acting effectively, and $q : \mathcal{P} \to \bar{\mathcal{P}}_0$ is a double covering.

As for the Galilei group in Definition **??**, it is convenient to name the transformations from which every element is built. The only difference is that the boosts are now parametrized by the rapidity vector $\mathbf{w}$, because velocities do not add linearly in special relativity, whereas rapidities, as we shall see, do for boosts along a fixed direction.

Definition 6.1.9 (Elementary transformations)**.** *Let $\mathcal{P}$ be the Poincaré group as per Definition 6.1.7. For $a \in \mathbb{R}^{1,3}$, $\mathbf{w} \in \mathbb{R}^3$ and $u \in SU(2)$ we set*

$$\tau_a := (a,\mathbb{1}), \qquad \beta_{\mathbf{w}} := \big(0, \exp\big(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\big)\big), \qquad \rho_u := (0,u), \tag{6.7}$$

and we call them the translation, the boost and the rotation. The time translation is $\tau_{(s,\mathbf{0})}$ and the spatial translation is $\tau_{(0,\mathbf{a})}$.

The boost $\beta_{\mathbf{w}}$ corresponds to a change to a frame moving with velocity $\tanh|\mathbf{w}|$ in the direction $\mathbf{w}/|\mathbf{w}|$, as the next lemma shows by computing its action on a time translation. The speed of light appears as the limit of infinite rapidity. The lemma also collects the relations between the elementary transformations, which are the counterpart of Lemma **??** and which are differentiated in Proposition 6.1.14.

Lemma 6.1.10 (Relations between the elementary transformations)**.** *Let $\mathcal{P}$ be as per Definition 6.1.7, with elementary transformations as per Definition 6.1.9. For all $a,a' \in \mathbb{R}^{1,3}$, $\mathbf{w} \in \mathbb{R}^3$, $s,s' \in \mathbb{R}$ and $u,u' \in SU(2)$ the following hold.*

1. *$\tau_a\tau_{a'} = \tau_{a+a'}$, $\rho_u\rho_{u'} = \rho_{uu'}$ and $\beta_{s\mathbf{w}}\beta_{s'\mathbf{w}} = \beta_{(s+s')\mathbf{w}}$.*
2. *$(0,g)\,\tau_a\,(0,g)^{-1} = \tau_{\Lambda(g)a}$ for every $g \in SL(2,\mathbb{C})$.*

3. $\rho_u \tau_a \rho_u^{-1} = \tau_{(a^0, R_u \mathbf{a})}$ *for* $a = (a^0, \mathbf{a})$*, and* $\rho_u \beta_{\mathbf{w}} \rho_u^{-1} = \beta_{R_u \mathbf{w}}$*, where* $R_u \in SO(3)$ *is the rotation determined by* u *through* $u\,(\mathbf{x}\cdot\boldsymbol{\sigma})\,u^{-1} = (R_u \mathbf{x})\cdot\boldsymbol{\sigma}$.

4. $\beta_{\mathbf{w}} \tau_{(s,\mathbf{0})} \beta_{\mathbf{w}}^{-1} = \tau_{(s\cosh|\mathbf{w}|,\; s\sinh|\mathbf{w}|\,\mathbf{w}/|\mathbf{w}|)}$ *for* $\mathbf{w} \neq \mathbf{0}$.

5. *Every element of* $\mathcal{P}$ *is written in a unique way as* $\tau_a \beta_{\mathbf{w}} \rho_u$*, and the map* $(a, \mathbf{w}, u) \mapsto \tau_a \beta_{\mathbf{w}} \rho_u$ *is a diffeomorphism* $\mathbb{R}^4 \times \mathbb{R}^3 \times SU(2) \to \mathcal{P}$.

Proof. Item 1. The first two relations follow from Definition 6.1.7, since $\Lambda(\mathbb{1}) = \mathrm{id}$ and $(0,u)(0,u') = (0,uu')$. For the third, $\exp\big(\tfrac{1}{2} s\,\mathbf{w}\cdot\boldsymbol{\sigma}\big)\exp\big(\tfrac{1}{2} s'\,\mathbf{w}\cdot\boldsymbol{\sigma}\big) = \exp\big(\tfrac{1}{2}(s+s')\,\mathbf{w}\cdot\boldsymbol{\sigma}\big)$ because the exponents commute.

Item 2. By Definition 6.1.7, $(0,g)(a,\mathbb{1}) = (\Lambda(g)a, g) = (\Lambda(g)a, \mathbb{1})(0,g)$.

Item 3. For $u \in SU(2)$ one has $\Lambda(u)(x^0, \mathbf{x}) = (x^0, R_u \mathbf{x})$, because $u\mathbb{1}u^{-1} = \mathbb{1}$ and $u(\mathbf{x}\cdot\boldsymbol{\sigma})u^{-1}$ is a traceless Hermitian matrix, hence of the form $\mathbf{y}\cdot\boldsymbol{\sigma}$. Since $\det(\mathbf{x}\cdot\boldsymbol{\sigma}) = -|\mathbf{x}|^2$ is preserved by conjugation, $|\mathbf{y}| = |\mathbf{x}|$, and $R_u : \mathbf{x} \mapsto \mathbf{y}$ is a linear isometry of $\mathbb{R}^3$. It has determinant $+1$ because $SU(2)$ is connected. The first relation is item 2. For the second, $u\exp\big(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\big)u^{-1} = \exp\big(\tfrac{1}{2} u(\mathbf{w}\cdot\boldsymbol{\sigma})u^{-1}\big) = \exp\big(\tfrac{1}{2}(R_u\mathbf{w})\cdot\boldsymbol{\sigma}\big)$.

Item 4. By item 2 the left-hand side is $\tau_{\Lambda(g)a}$ with $g = \exp\big(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\big)$ and $a = (s, \mathbf{0})$. The matrix g is Hermitian, so $\Lambda(g)a$ is represented by $g\,(s\mathbb{1})\,g = s\,e^{\mathbf{w}\cdot\boldsymbol{\sigma}} = s\big(\cosh|\mathbf{w}|\,\mathbb{1} + \sinh|\mathbf{w}|\,\frac{\mathbf{w}}{|\mathbf{w}|}\cdot\boldsymbol{\sigma}\big)$, which is the stated vector.

Item 5. By Definition 6.1.7, $\tau_a\beta_{\mathbf{w}}\rho_u = (a, \exp\big(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\big)u) = (a, \Psi(\mathbf{w}, u))$ with Ψ as per Lemma 6.1.3, which is a diffeomorphism. □

Item 1 holds for boosts only along a fixed direction, since the product of two boosts with different directions is a boost composed with a rotation. Both statements are made concrete in the two examples below.

Example 6.1.11 (Collinear boosts and the addition of velocities). Take $\mathbf{w} = w\,\mathbf{e}_3$. Then $\exp\big(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\big) = \mathrm{diag}(e^{w/2}, e^{-w/2})$ and Equation (6.3) gives $x^0 \pm x^3 \mapsto e^{\pm w}(x^0 \pm x^3)$, with x^1 and x^2 unchanged. This is the hyperbolic rotation

$$x'^0 = x^0\cosh w + x^3 \sinh w, \qquad x'^3 = x^0 \sinh w + x^3 \cosh w,$$

in the plane (x^0, x^3). The rapidities of collinear boosts add by item 1 of Lemma 6.1.10, and since the velocity of a boost of rapidity w is $v = \tanh w$, the velocity of the composition is $\tanh\big(w + w'\big) = (v + v')/(1 + vv')$. This is the relativistic addition of velocities, and it shows that the velocity of the composition never exceeds one. In the Galilean group boosts compose by $\beta_{\mathbf{v}}\beta_{\mathbf{v}'} = \beta_{\mathbf{v}+\mathbf{v}'}$ in all directions.

Example 6.1.12 (Perpendicular boosts and the Thomas–Wigner rotation). Let $c_i := \cosh(w_i/2)$, $s_i := \sinh(w_i/2)$, and consider the product $g = e^{\frac{1}{2}w_1\sigma_1} e^{\frac{1}{2}w_2\sigma_2}$ of two boosts along perpendicular directions. Since $\sigma_1\sigma_2 = i\sigma_3$,

$$g = c_1c_2\,\mathbb{1} + s_1c_2\,\sigma_1 + c_1s_2\,\sigma_2 + i\,s_1s_2\,\sigma_3.$$

This matrix is not Hermitian, so g is not a pure boost. Its polar decomposition $g = HU$ of Lemma 6.1.3 is obtained by choosing $U = \cos\frac{\vartheta}{2}\,\mathbb{1} + i\sin\frac{\vartheta}{2}\,\sigma_3$ and requiring that gU^{-1} be Hermitian. The coefficient of σ_3 in gU^{-1} is $i(s_1s_2\cos\frac{\vartheta}{2} - c_1c_2\sin\frac{\vartheta}{2})$, which vanishes if and only if

$$\tan\frac{\vartheta}{2} = \tanh\frac{w_1}{2}\,\tanh\frac{w_2}{2},$$

and then all the other coefficients of gU^{-1} are real and its scalar part is positive, so that $H = gU^{-1}$ is positive definite. Hence the composition of two perpendicular boosts is a pure boost followed by a rotation about the axis orthogonal to their plane, by an angle of modulus $\vartheta \simeq w_1w_2/2$ for small rapidities. This is the Thomas rotation, and it is the elementary reason why a relativistic particle cannot be given a spin which is independent of its momentum.

The relations of Lemma 6.1.10 should be compared with those of Lemma **??**. The comparison identifies where the Poincaré group differs from the Galilei group at the level of the group law, before any representation is considered.

Remark 6.1.13 (Comparison with the Galilei group). In the Galilei group, a boost and a spatial translation commute, whereas the lifts of Lemma **??** commute only up to the central element $z(\mathbf{v}\cdot\mathbf{a})$. In the Poincaré group no central element is available, and the interplay between boosts and translations is encoded in item 2 of Lemma 6.1.10. A boost does not commute with a spatial translation along its direction, since it maps it to a translation with a nonvanishing time component, and it does not commute with a time translation, since by item 4 it maps it to a translation with a spatial component. The Galilean relation $\hat\tau_{-s}\hat\beta_{\mathbf{v}}\hat\tau_s = z(-\frac{1}{2}s|\mathbf{v}|^2)\hat\beta_{\mathbf{v}}\hat\alpha_{s\mathbf{v}}$ of Lemma **??** has the same content as item 4, with a conjugation of the boost by a time translation instead of a conjugation of the time translation by a boost, and the phase present there is the one that the central extension of Section **??** was introduced to absorb. Moreover boosts with different directions do not compose additively, and their product contains a rotation, which is the Thomas rotation of Example 6.1.12. This also has no counterpart in the Galilei group, where $\beta_{\mathbf{v}}\beta_{\mathbf{v}'} = \beta_{\mathbf{v}+\mathbf{v}'}$.

The Lie Algebra

To pass from these relations to the generators of the unitary representations of Section 6.3, and to decide whether the group has central extensions, we need the commutation relations of the Lie algebra of $\mathcal{P}$. They are obtained by differentiating the group law, and the cleanest way to do so is to realize $\mathcal{P}$ as a matrix group. The matrices $r_j = -\frac{i}{2}\sigma_j$ below are the generators T_j of Example 1.2.5, while the boost generators b_j are new.

Proposition 6.1.14 (Lie algebra of the Poincaré group)**.** *Let ρ be as in Equation* (6.4)*. The Lie algebra $\mathfrak{p}$ of $\mathcal{P}$ is the real vector space $\mathfrak{sl}(2,\mathbb{C})\oplus\mathbb{R}^{1,3}$ with the bracket*

$$[(\xi,x),(\eta,y)] = \big([\xi,\eta],\ \rho(\xi)y - \rho(\eta)x\big). \tag{6.8}$$

Let $r_j := (-\frac{i}{2}\sigma_j, 0)$, $b_j := (\frac{1}{2}\sigma_j, 0)$ for $j = 1,2,3$ and $t_\mu := (0,\mathbf{e}_\mu)$ for $\mu = 0,\dots,3$, where $\mathbf{e}_\mu$ is the canonical basis of $\mathbb{R}^{1,3}$. Then these ten elements form a basis of $\mathfrak{p}$, and up to antisymmetry the only nonvanishing brackets between them are

$$[r_j,r_k] = \varepsilon_{jkl}\,r_l, \qquad [r_j,b_k] = \varepsilon_{jkl}\,b_l, \qquad [b_j,b_k] = -\varepsilon_{jkl}\,r_l, \tag{6.9}$$

$$[r_j,t_k] = \varepsilon_{jkl}\,t_l, \qquad [b_j,t_0] = t_j, \qquad [b_j,t_k] = \delta_{jk}\,t_0, \tag{6.10}$$

with ε_{jkl} the Levi-Civita symbol and a sum over the repeated index l.

Proof. Consider the smooth map $F : \mathcal{P} \to GL(5,\mathbb{R})$ given in block form by

$$F(a,g) = \begin{pmatrix} \Lambda(g) & a \\ 0 & 1 \end{pmatrix},$$

which is a homomorphism because $F(a,g)F(a',g') = \begin{pmatrix} \Lambda(g)\Lambda(g') & \Lambda(g)a'+a \\ 0 & 1 \end{pmatrix} = F\big((a,g)(a',g')\big)$. Along the curve $t \mapsto (tx, \exp(t\xi))$ one has $\Lambda(\exp(t\xi)) = \exp(t\rho(\xi))$, by the naturality property in Proposition 1.2.14 applied to the homomorphism of Lemma 6.1.5. The differential of F at the identity is therefore

$$dF(\xi,x) = \begin{pmatrix} \rho(\xi) & x \\ 0 & 0 \end{pmatrix}.$$

This map is injective by Lemma 6.1.5, and it is a Lie algebra homomorphism into $\mathfrak{gl}(5,\mathbb{R}) = M_5(\mathbb{R})$ with the commutator of Example 1.2.4, since the differential of a Lie group homomorphism preserves the brackets [**TODO-Lee13**]. A direct computation of the commutator of two such matrices yields

$$\left[\begin{pmatrix} \rho(\xi) & x \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} \rho(\eta) & y \\ 0 & 0 \end{pmatrix}\right] = \begin{pmatrix} \rho([\xi,\eta]) & \rho(\xi)y - \rho(\eta)x \\ 0 & 0 \end{pmatrix},$$

where $[\rho(\xi),\rho(\eta)] = \rho([\xi,\eta])$ because ρ is the differential of a homomorphism. Since dF is an injective homomorphism, Equation (6.8) follows.

The elements r_j and b_j form a basis of $\mathfrak{sl}(2,\mathbb{C})$ as a real vector space, because $\frac{1}{2}\sigma_j$ and $\frac{i}{2}\sigma_j$ span the traceless complex matrices. For the brackets in $\mathfrak{sl}(2,\mathbb{C})$, $[\sigma_j,\sigma_k]=2i\varepsilon_{jkl}\sigma_l$ yields $[r_j,r_k]=-\frac{1}{4}[\sigma_j,\sigma_k]=-\frac{i}{2}\varepsilon_{jkl}\sigma_l=\varepsilon_{jkl}r_l$, then $[r_j,b_k]=-\frac{i}{4}[\sigma_j,\sigma_k]=\frac{1}{2}\varepsilon_{jkl}\sigma_l=\varepsilon_{jkl}b_l$ and $[b_j,b_k]=\frac{1}{4}[\sigma_j,\sigma_k]=\frac{i}{2}\varepsilon_{jkl}\sigma_l=-\varepsilon_{jkl}r_l$. For the action on translations we evaluate ρ on $x=(x^0,\mathbf{x})$. Since $r_j^\dagger=-r_j$, $\widehat{\rho(r_j)x}=[r_j,\hat{x}]=\varepsilon_{jkl}x^k\sigma_l$, so that $\rho(r_j)x=(0,\mathbf{e}_j\times\mathbf{x})$. Since $b_j^\dagger=b_j$, $\widehat{\rho(b_j)x}=\frac{1}{2}\{\sigma_j,\hat{x}\}=x^0\sigma_j+x^j\mathbb{1}$, so that $\rho(b_j)x=(x^j,\,x^0\mathbf{e}_j)$. Equation (6.8) yields $[(\xi,0),(0,x)]=(0,\rho(\xi)x)$ and $[(0,x),(0,y)]=0$, and evaluating on $x=\mathbf{e}_0$ and $x=\mathbf{e}_k$ yields Equation (6.10). □

The brackets can be read as a dictionary. The first line of Equation (6.9) says that rotations close on themselves. The second says that boosts form a vector under rotations. The third, $[b_j,b_k]=-\varepsilon_{jkl}r_l$, is the infinitesimal form of the Thomas rotation of Example 6.1.12: two boosts in different directions do not commute, and their commutator is a rotation. In the Galilei algebra this bracket vanishes. The second line, Equation (6.10), says that rotations rotate translations, that a boost turns a time translation into a spatial one, which is the infinitesimal form of item 4 of Lemma 6.1.10, and that a boost turns a spatial translation into a time translation, $[b_j,t_k]=\delta_{jk}t_0$. This last bracket has no counterpart in the Galilei algebra, where the corresponding commutator is central, and Section 6.2 shows that this is exactly why the Poincaré algebra has no central extension.

The Chiral Decomposition of the Lorentz Algebra

The Lorentz algebra has a remarkable structure that is invisible over the reals and appears as soon as we complexify: rotations and boosts combine into two commuting copies of $\mathfrak{su}(2)$. We need this fact twice, to prove that the Lorentz algebra is semisimple, which is the hypothesis of Whitehead's lemmas in Section 6.2, and to classify its finite-dimensional representations in Section 6.5. We write $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{C}}:=\mathfrak{sl}(2,\mathbb{C})\otimes_{\mathbb{R}}\mathbb{C}$. For brevity we write r_j and b_j also for the corresponding elements $(\xi,0)$ of $\mathfrak{sl}(2,\mathbb{C})$.

Lemma 6.1.15 (Chiral decomposition)**.** *In $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{C}}$ let $A_k:=\frac{1}{2}(r_k-ib_k)$ and $B_k:=\frac{1}{2}(r_k+ib_k)$. Then*

$$[A_j,A_k]=\sum_l\varepsilon_{jkl}A_l,\qquad [B_j,B_k]=\sum_l\varepsilon_{jkl}B_l,\qquad [A_j,B_k]=0, \tag{6.11}$$

and $r_k=A_k+B_k$, $b_k=i(A_k-B_k)$. Hence $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{C}}\cong\mathfrak{su}(2)_{\mathbb{C}}\oplus\mathfrak{su}(2)_{\mathbb{C}}$, the sum of the two commuting ideals spanned by the A_k and by the B_k.

Proof. By Equation (6.9), and using $[b_j,r_k]=\sum_l\varepsilon_{jkl}b_l$,

$$[A_j,A_k]=\tfrac{1}{4}\big([r_j,r_k]-i[r_j,b_k]-i[b_j,r_k]-[b_j,b_k]\big)=\tfrac{1}{4}\sum_l\varepsilon_{jkl}\big(2r_l-2ib_l\big)=\sum_l\varepsilon_{jkl}A_l,$$
$$[A_j,B_k]=\tfrac{1}{4}\big([r_j,r_k]+i[r_j,b_k]-i[b_j,r_k]+[b_j,b_k]\big)=\tfrac{1}{4}\sum_l\varepsilon_{jkl}\big(r_l+ib_l-ib_l-r_l\big)=0,$$

and the computation for B is the complex conjugate of the first one. The expressions of r_k and b_k are solved from the definitions. □

Corollary 6.1.16 (Semisimplicity and perfectness)**.** 1. *The real Lie algebra $\mathfrak{l}:=\mathfrak{sl}(2,\mathbb{C})$ is semisimple.*

2. *$[\mathfrak{p},\mathfrak{p}]=\mathfrak{p}$, i.e., the Poincaré algebra is perfect.*

Proof. Item 1. Each ideal of Lemma 6.1.15 is isomorphic to $\mathfrak{su}(2)_{\mathbb{C}}=\mathfrak{sl}(2,\mathbb{C})$ as a complex Lie algebra, which is simple. Hence the complexification of $\mathfrak{l}$ is semisimple, and a real Lie algebra is semisimple if and only if its complexification is [**TODO-Hum10**]. Item 2. By Equations (6.9) and (6.10), $r_l=[r_j,r_k]$ and $b_l=[r_j,b_k]$ for (j,k,l) cyclic, $t_j=[b_j,t_0]$ and $t_0=[b_1,t_1]$. □

Item 2 will be used in Proposition 6.3.5 to show that no phase ambiguity survives in the lift of a Poincaré action. We conclude the section with the topology of $\mathcal{P}$, which is what makes Bargmann's theorem applicable in Section 6.2.

Corollary 6.1.17 (Topology of the Poincaré group)**.** *The group $\mathcal{P}$ is diffeomorphic to $\mathbb{R}^7 \times S^3$, hence it is a connected and simply connected Lie group of dimension* 10*. The homomorphism $q : \mathcal{P} \to \bar{\mathcal{P}}_0$ of Remark 6.1.8 is the universal covering of the proper orthochronous Poincaré group, and $\pi_1(SO^\uparrow(1,3)) \cong \mathbb{Z}_2$.*

Proof. By item 5 of Lemma 6.1.10, $\mathcal{P} \simeq \mathbb{R}^4 \times \mathbb{R}^3 \times SU(2)$, and $SU(2) \simeq S^3$ is simply connected. The map q is a covering map with kernel $\{(0, \pm\mathbb{1})\}$ by Lemma 6.1.5 and Remark 6.1.8, and its total space is simply connected, so it is the universal covering. In a universal covering the group of deck transformations is isomorphic to the fundamental group of the base, and the deck group of $SL(2,\mathbb{C}) \to SO^\uparrow(1,3)$ is $\{\pm\mathbb{1}\} \cong \mathbb{Z}_2$. □

The fundamental group $\mathbb{Z}_2$ is the topological source of the sign ambiguity discussed in Example **??**, and Section 6.2 shows that it is the only obstruction left.

6.2 Absence of Central Extensions

A symmetry acts on rays, and the unitary operators that implement it are determined only up to a phase. When the group law is imposed on such operators, the phases may or may not be removable. If they are not, the group has to be enlarged by a central element, and this is exactly what happened for the Galilei group: Section **??** showed that $H^2(\mathfrak{g}, \mathbb{R}) \cong \mathbb{R}$ by Proposition **??**, and the mass entered the theory as the class of a cocycle and, by Theorem **??**, as the parameter of the central extension that turns a projective representation into a genuine one. The question we ask now is whether anything of this kind happens for the Poincaré group.

The answer is no, and the reason is a rigidity property of the Lorentz algebra, which we make explicit. At the infinitesimal level a central extension is a cocycle: a bilinear antisymmetric form ψ on the Lie algebra, satisfying the Jacobi-type identity of Proposition **??**, that records the phase picked up by the commutator of two generators. It is harmless if it is a coboundary, because then it can be absorbed by shifting each generator X by a multiple $\lambda(X)\mathbb{1}$ of the identity, and it is relevant otherwise. For a semisimple Lie algebra there is always enough room for such shifts, which is the content of Whitehead's lemmas. The Poincaré algebra is not semisimple, since the translations form an abelian ideal, but it is the semidirect sum of a semisimple algebra, the Lorentz algebra, with an abelian one, and its cocycles can be peeled off in layers. The Lorentz layer and the mixed layer are removed by Whitehead's lemmas. What is left is a form on the translations, which is killed by invariance under rotations.

We prove that $H^2(\mathfrak{p}, \mathbb{R}) = 0$ for the Lie algebra of Proposition 6.1.14. The consequence, through Proposition **??**, is that every continuous projective unitary representation of $\mathcal{P}$ lifts to a genuine one without enlarging the group. This changes the origin of the mass, which can no longer be read from a phase and will be recovered in Section 6.3.3 as the eigenvalue of a Casimir operator. A residual sign survives, and Proposition 6.2.5 shows that it is the topological obstruction of Example **??**.

Throughout the section $\mathfrak{l} := \mathfrak{sl}(2,\mathbb{C})$, regarded as a real Lie algebra, and $\mathfrak{t} := \mathbb{R}^{1,3}$, so that $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{t}$ as in Proposition 6.1.14. We identify $\xi \in \mathfrak{l}$ with $(\xi, 0)$ and $x \in \mathfrak{t}$ with $(0, x)$, so that Equation (6.8) reads

$$[\xi, \eta] = [\xi, \eta]_\mathfrak{l}, \qquad [\xi, x] = \rho(\xi)x, \qquad [x, y] = 0, \tag{6.12}$$

for $\xi, \eta \in \mathfrak{l}$ and $x, y \in \mathfrak{t}$. By Proposition **??**, a 2-cocycle of $\mathfrak{p}$ is a bilinear antisymmetric map $\psi : \mathfrak{p} \times \mathfrak{p} \to \mathbb{R}$ with $\psi([X,Y],Z) + \psi([Y,Z],X) + \psi([Z,X],Y) = 0$, and the coboundaries are the maps $\psi_\lambda(X,Y) = \lambda([X,Y])$ with $\lambda \in \mathfrak{p}^*$.

Two Classical Tools

The first two steps of the proof rely on two classical results on semisimple Lie algebras. The second Whitehead lemma was already used in Example **??** for $\mathfrak{su}(2)$, and we need it here for $\mathfrak{l}$ together with its first counterpart, which involves cocycles with values in a representation. In words, the first lemma says that a map ϕ from a semisimple algebra to a module, obeying the rule satisfied by the derivative of $\xi \mapsto \varrho(\xi)m$, is of that form: every "infinitesimal" deformation is trivial.

Theorem 6.1 (Whitehead's lemmas)**.** *Let $\mathfrak{g}$ be a finite dimensional semisimple real Lie algebra and let $\varrho : \mathfrak{g} \to \mathfrak{gl}(M)$ be a representation on a finite dimensional real vector space M.*

1. *If a linear map $\phi : \mathfrak{g} \to M$ satisfies $\phi([\xi,\eta]) = \varrho(\xi)\phi(\eta) - \varrho(\eta)\phi(\xi)$ for all $\xi,\eta \in \mathfrak{g}$, then there is $m \in M$ such that $\phi(\xi) = \varrho(\xi)m$ for all $\xi \in \mathfrak{g}$.*

2. *$H^2(\mathfrak{g},\mathbb{R}) = 0$, i.e., every 2-cocycle ψ of $\mathfrak{g}$ is of the form ψ_λ for some $\lambda \in \mathfrak{g}^*$.*

See [**TODO-Hum10**], where the statements are proved over $\mathbb{C}$, and they hold for real Lie algebras since the cohomologies involved are compatible with complexification.

We apply Theorem 6.1 to the Lorentz algebra. Its semisimplicity was established in Corollary 6.1.16, through the chiral decomposition, because the algebra is regarded as a real one. The next remark identifies the representation of $\mathfrak{l}$ on which the first lemma will be used.

Remark 6.2.1. The representation of $\mathfrak{l}$ on $\mathfrak{t}$ is ρ of Equation (6.4), and the dual space $\mathfrak{t}^*$ carries the dual representation $\varrho(\xi)f := -f \circ \rho(\xi)$. One checks that $[\varrho(\xi),\varrho(\eta)]f = -f \circ \rho([\xi,\eta]) = \varrho([\xi,\eta])f$, so that ϱ is a representation, and both ρ and ϱ are finite dimensional. Hence Theorem 6.1 applies to $\mathfrak{g} = \mathfrak{l}$ with $M = \mathfrak{t}^*$, by item 1 of Corollary 6.1.16.

The third step does not use any semisimplicity. After the first two steps the only surviving part of the cocycle is an antisymmetric form on the translations that is invariant under the Lorentz algebra, and it suffices to use the rotations. Intuitively, a form on $\mathbb{R} \oplus \mathbb{R}^3$ that is invariant under rotations must be built from a rotation-invariant vector or from the cross product of two vectors, and there is no invariant vector in $\mathbb{R}^3$.

Lemma 6.2.2 (Rotation invariant forms)**.** *Let $\omega : \mathfrak{t} \times \mathfrak{t} \to \mathbb{R}$ be a bilinear antisymmetric map such that $\omega(\rho(r_j)x, y) + \omega(x, \rho(r_j)y) = 0$ for all $x, y \in \mathfrak{t}$ and $j = 1,2,3$. Then $\omega = 0$.*

Proof. Every bilinear antisymmetric map on $\mathbb{R}^{1,3}$ is of the form

$$\omega(x,y) = x^0\,\mathbf{v}\cdot\mathbf{y} - y^0\,\mathbf{v}\cdot\mathbf{x} + \mathbf{k}\cdot(\mathbf{x}\times\mathbf{y})$$

for unique $\mathbf{v},\mathbf{k} \in \mathbb{R}^3$, namely $v^j = \omega(\mathbf{e}_0,\mathbf{e}_j)$ and $k^l = \frac{1}{2}\varepsilon_{jkl}\,\omega(\mathbf{e}_j,\mathbf{e}_k)$. By the proof of Proposition 6.1.14, $\rho(r_j)x = (0, \mathbf{e}_j \times \mathbf{x})$. For $\mathbf{n} = \sum_j n^j \mathbf{e}_j$ the linear combination of the invariance conditions reads $\omega(\mathbf{n} \times \mathbf{x}, y) + \omega(x, \mathbf{n} \times \mathbf{y}) = 0$, where $\mathbf{n} \times \mathbf{x}$ stands for $(0, \mathbf{n} \times \mathbf{x})$. Taking $\mathbf{x} = \mathbf{0}$ yields $x^0\,\mathbf{v}\cdot(\mathbf{n}\times\mathbf{y}) = 0$ for all x^0, $\mathbf{y}$ and $\mathbf{n}$, so that $\mathbf{n}\cdot(\mathbf{y}\times\mathbf{v}) = 0$ for all $\mathbf{n}$ and $\mathbf{y}$. Hence $\mathbf{y}\times\mathbf{v} = \mathbf{0}$ for all $\mathbf{y}$, which entails $\mathbf{v} = \mathbf{0}$. With $\mathbf{v} = \mathbf{0}$ the condition becomes

$$\mathbf{k}\cdot\big((\mathbf{n}\times\mathbf{x})\times\mathbf{y} + \mathbf{x}\times(\mathbf{n}\times\mathbf{y})\big) = \mathbf{k}\cdot\big(\mathbf{n}\times(\mathbf{x}\times\mathbf{y})\big) = \mathbf{n}\cdot\big((\mathbf{x}\times\mathbf{y})\times\mathbf{k}\big) = 0,$$

where the first equality is the Jacobi identity of the cross product. Since $\mathbf{n}$ is arbitrary, $(\mathbf{x}\times\mathbf{y})\times\mathbf{k} = \mathbf{0}$ for all $\mathbf{x},\mathbf{y}$, and choosing $\mathbf{x}\times\mathbf{y}$ orthogonal to $\mathbf{k}$ and nonvanishing, if $\mathbf{k} \neq \mathbf{0}$, yields a contradiction. Hence $\mathbf{k} = \mathbf{0}$. □

The lemma holds because the rotation group has no nonzero invariant vector in $\mathbb{R}^3$, and the space of antisymmetric forms on $\mathbb{R} \oplus \mathbb{R}^3$ is isomorphic to two copies of $\mathbb{R}^3$.

The Vanishing of H^2

We have now all the ingredients to prove the vanishing of the second cohomology group of $\mathfrak{p}$. The argument is a decomposition along $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{t}$, in which the three components of a cocycle, the Lorentz one, the mixed one and the translation one, are removed one at a time by subtracting coboundaries. Each subtraction is allowed because $\lambda \mapsto \psi_\lambda$ is linear and every ψ_λ is a cocycle, so that the difference of a cocycle and a coboundary is still a cocycle.

Theorem 6.2 (H^2 of the Poincaré algebra)**.** *Let $\mathfrak{p}$ be the Lie algebra of $\mathcal{P}$. Then $H^2(\mathfrak{p},\mathbb{R}) = 0$, i.e., every 2-cocycle ψ of $\mathfrak{p}$ is of the form ψ_λ for some $\lambda \in \mathfrak{p}^*$.*

Proof. Let ψ be a 2-cocycle of $\mathfrak{p}$.

Step 1, the Lorentz sector. The restriction of ψ to $\mathfrak{l} \times \mathfrak{l}$ is a 2-cocycle of $\mathfrak{l}$, because $\mathfrak{l}$ is a subalgebra. By item 1 of Corollary 6.1.16 and item 2 of Theorem 6.1 there is $\lambda_0 \in \mathfrak{l}^*$ with $\psi(\xi,\eta) = \lambda_0([\xi,\eta])$ for $\xi,\eta \in \mathfrak{l}$.

We extend λ_0 to $\mathfrak{p}$ by $\lambda_0|_{\mathfrak{t}} = 0$. Equation (6.12) yields $\psi_{\lambda_0}(\xi, x) = \lambda_0(\rho(\xi)x) = 0$ and $\psi_{\lambda_0}(x, y) = 0$, so that $\psi' := \psi - \psi_{\lambda_0}$ is a cocycle vanishing on $\mathfrak{l} \times \mathfrak{l}$ and satisfying $\psi'(\xi, x) = \psi(\xi, x)$ and $\psi'(x, y) = \psi(x, y)$.

Step 2, the mixed sector. Set $\phi(\xi)(x) := \psi'(\xi, x)$, which defines a linear map $\phi : \mathfrak{l} \to \mathfrak{t}^*$. The cocycle condition for ψ' on (ξ, η, x) with $\xi, \eta \in \mathfrak{l}$ and $x \in \mathfrak{t}$ reads

$$\psi'([\xi, \eta], x) + \psi'([\eta, x], \xi) + \psi'([x, \xi], \eta) = 0.$$

By Equation (6.12) and the antisymmetry of ψ', the first term is $\phi([\xi, \eta])(x)$, the second is $-\phi(\xi)(\rho(\eta)x)$ and the third is $\phi(\eta)(\rho(\xi)x)$. Therefore

$$\phi([\xi, \eta])(x) = \phi(\xi)(\rho(\eta)x) - \phi(\eta)(\rho(\xi)x).$$

In terms of the dual representation ϱ of Remark 6.2.1, $\phi(\xi) \circ \rho(\eta) = -\varrho(\eta)\phi(\xi)$, and this identity becomes $\phi([\xi, \eta]) = \varrho(\xi)\phi(\eta) - \varrho(\eta)\phi(\xi)$. Item 1 of Theorem 6.1 yields $m \in \mathfrak{t}^*$ with $\phi(\xi) = \varrho(\xi)m = -m \circ \rho(\xi)$, *i.e.*, $\psi'(\xi, x) = -m(\rho(\xi)x)$. Let $\mu \in \mathfrak{p}^*$ be the extension of m with $\mu|_{\mathfrak{l}} = 0$. Then $\psi_\mu(\xi, x) = \mu(\rho(\xi)x) = m(\rho(\xi)x)$, while $\psi_\mu(\xi, \eta) = \mu([\xi, \eta]_{\mathfrak{l}}) = 0$ and $\psi_\mu(x, y) = 0$. It descends that $\psi'' := \psi' + \psi_\mu$ is a cocycle vanishing on $\mathfrak{l} \times \mathfrak{l}$ and on $\mathfrak{l} \times \mathfrak{t}$.

Step 3, the translation sector. The cocycle condition for ψ'' on (ξ, x, y) with $\xi \in \mathfrak{l}$ and $x, y \in \mathfrak{t}$ reads $\psi''([\xi, x], y) + \psi''([x, y], \xi) + \psi''([y, \xi], x) = 0$. The second term vanishes because $[x, y] = 0$, and the third equals $\psi''(x, \rho(\xi)y)$ by antisymmetry, so that

$$\psi''(\rho(\xi)x, y) + \psi''(x, \rho(\xi)y) = 0.$$

In particular this holds for $\xi = r_j$, and Lemma 6.2.2 yields $\psi''|_{\mathfrak{t}\times\mathfrak{t}} = 0$. Since ψ'' vanishes on all three sectors, $\psi'' = 0$ by bilinearity and antisymmetry. Collecting the three steps, $\psi = \psi_{\lambda_0} - \psi_\mu = \psi_{\lambda_0 - \mu}$. □

The semisimplicity of $\mathfrak{l}$ is used only in Steps 1 and 2 through Theorem 6.1, while Step 3 uses the rotations alone.

The vanishing of $H^2(\mathfrak{p}, \mathbb{R})$ contrasts with the Galilei algebra, and the precise point of the contrast deserves to be spelled out. The Galilean cocycle is a boost-translation pairing, and it is available because the corresponding bracket vanishes in the Galilei algebra: a pairing that is nonzero on a pair of commuting generators cannot be a coboundary.

Remark 6.2.3 (Comparison with Proposition **??**). The generator of $H^2(\mathfrak{g}, \mathbb{R})$ for the Galilei algebra is the class of ψ_ζ, with $\psi_\zeta(X_i^\beta, X_j^\alpha) = \delta_{ij}$ by Proposition **??**. This value cannot be a coboundary, because $[X_i^\beta, X_j^\alpha] = 0$ entails $\psi_\lambda(X_i^\beta, X_j^\alpha) = 0$ for every $\lambda \in \mathfrak{g}^*$. In the Poincaré algebra the corresponding bracket is $[b_j, t_k] = \delta_{jk} t_0$ by Equation (6.10), and it does not vanish. A pairing between boosts and translations with value δ_{jk} is then the coboundary ψ_λ with $\lambda(t_0) = 1$ and λ vanishing on the remaining basis elements, since $\psi_\lambda(b_j, t_k) = \lambda(\delta_{jk} t_0) = \delta_{jk}$. The nonvanishing bracket between boosts and translations is therefore what removes the Galilean cocycle. Physically, the phase $e^{im\mathbf{v}\cdot\mathbf{a}/\hbar}$ between a boost and a translation can be absorbed by redefining the energy by a constant, which in the relativistic algebra is the generator t_0 of time translations itself. The vanishing of H^2 depends on the number of space dimensions, as Remark 6.2.6 shows.

Consequences for Unitary Representations

We now translate the algebraic statement into the language of unitary representations. The group $\mathcal{P}$ is connected and simply connected by Corollary 6.1.17, so Bargmann's correspondence of Proposition **??** applies, and the argument follows the proof of Theorem **??** with the cocycle $\zeta = 0$.

Corollary 6.2.4 (Lift of projective representations of $\mathcal{P}$)**.** *Let $\mathcal{H}$ be a separable Hilbert space and let $\kappa \mapsto V_\kappa$ be a continuous homomorphism from $\mathcal{P}$ to the group $U(\mathcal{H})/U(1)$ of unitary ray transformations, with the topology of Theorem* **??**. *Then there is a strongly continuous unitary representation $\Pi : \mathcal{P} \to U(\mathcal{H})$ such that $\Pi(\kappa)$ belongs to the ray V_κ for all $\kappa \in \mathcal{P}$.*

Proof. By items 1 and 2 of Proposition **??**, the class of the multiplier of V corresponds to an element of $H^2(\mathfrak{p},\mathbb{R})$, which vanishes by Theorem 6.2. The class of the multiplier is therefore the class of the trivial multiplier. Item 3 of the same proposition, applied to the continuous real cocycle $\zeta = 0$, shows that the phases of V can be chosen so that V is strongly continuous with multiplier exactly $e^{i0} = 1$, *i.e.*, that $\Pi(\kappa) := V_\kappa$ is a strongly continuous unitary representation of $\mathcal{P}$ with $\Pi(\kappa)$ in the ray of the original operator. □

The Galilean counterpart, Theorem **??**, yields a representation of the extension $\widehat{\mathcal{G}}$ and a parameter m, while here the lift is a representation of $\mathcal{P}$ itself.

Corollary 6.2.4 concerns the universal covering. The next step is to show that the passage from $\bar{\mathcal{P}}_0$ to $\mathcal{P}$ cannot be avoided in general, by computing the value of the central element $(0,-\mathbb{1})$ in an irreducible representation. This is the relativistic version of Example **??**, where the sign was produced by a rotation by 2π.

Proposition 6.2.5 (The central sign)**.** *Let $z := (0,-\mathbb{1}) \in \mathcal{P}$. Then z is central, $z^2 = e$ for the identity e of $\mathcal{P}$, and in every irreducible strongly continuous unitary representation Π of $\mathcal{P}$ there is $s_\Pi \in \{1,-1\}$ with $\Pi(z) = s_\Pi\,\mathbb{1}$.*

Proof. By Lemma 6.1.5, $\Lambda(-\mathbb{1}) = \mathrm{id}$, so $z(a,g) = (a,-g) = (a,g)z$ and z is central, while $z^2 = (0,\mathbb{1})$. The operator $\Pi(z)$ is unitary with $\Pi(z)^2 = \mathbb{1}$, so $\Pi(z)^* = \Pi(z)^{-1} = \Pi(z)$ and $\Pi(z)$ is selfadjoint. It commutes with the set $\mathcal{M} = \Pi(\mathcal{P})$, which is irreducible by assumption, and $e^{it\Pi(z)} = \cos t\,\mathbb{1} + i\sin t\,\Pi(z)$ commutes with $\mathcal{M}$ as well. Lemma **??** yields $\Pi(z) = s_\Pi\,\mathbb{1}$ with $s_\Pi \in \mathbb{R}$, and $s_\Pi^2 = 1$. □

If $s_\Pi = 1$ the representation Π factors through $\bar{\mathcal{P}}_0$, while if $s_\Pi = -1$ only the associated ray representation does. The value of s_Π is computed on the representations of Section 6.4, where it is shown to be $(-1)^{2j}$ (Corollary 6.4.10): $s_\Pi = -1$ corresponds to half integer spin.

The vanishing of H^2 relies on the semisimplicity of the Lorentz algebra, and the last remark shows that the statement fails when this hypothesis does not hold. It also fixes the range of validity of Theorem 6.2.

Remark 6.2.6 (The case of one space dimension). Theorem 6.2 is a statement on $\mathbb{R}^{1,3}$. In one space dimension the Poincaré algebra has basis k, t_0, t_1 with $[k,t_0] = t_1$, $[k,t_1] = t_0$ and $[t_0,t_1] = 0$, and the Lorentz algebra spanned by k is abelian, hence not semisimple. The bilinear antisymmetric map ψ with $\psi(t_0,t_1) = 1$ and all other independent values equal to zero is a cocycle, since the cocycle condition on (k,t_0,t_1) reads $\psi([k,t_0],t_1) + \psi([t_0,t_1],k) + \psi([t_1,k],t_0) = \psi(t_1,t_1) - \psi(t_0,t_0) = 0$, and on the remaining triples it holds by antisymmetry and $[t_0,t_1] = 0$. It is not a coboundary, because $\psi_\lambda(t_0,t_1) = \lambda([t_0,t_1]) = 0$ for every λ. Hence the Poincaré algebra in $1+1$ dimensions has a nontrivial central extension, and the argument of this section does not extend to it. In three space dimensions the rotations are non-abelian and semisimple, and this is what Steps 1 to 3 exploit.

6.3 Poincaré Invariance of the Projective State Space

The Poincaré group of Section 6.1 and the vanishing of its second cohomology group, proved in Section 6.2, are statements on the group alone. This section brings them to bear on the state space. The starting point is the one of Chapter **??**. The pure states are the points of the projective Hilbert space of Chapter **??**, which carries the Kähler structure of Equation (**??**), and a symmetry is a Kähler isomorphism of this manifold. The Hilbert space $\mathcal{H}$ underlying $\mathbb{P}\mathcal{H}$ is not assumed to be realized as a space of functions, and no operator on it is distinguished in advance. Consequently the section has to produce from the action of $\mathcal{P}$ alone the objects which textbook presentations postulate, namely the four-momentum, the mass and the energy.

The argument has four steps. We first state the setting and we record what is assumed and what is not. We then lift the action to a unitary representation, which is unique by the cohomological computation of Section 6.2, and the translations yield a spectral measure, which is the four-momentum. Irreducibility and positivity of the energy force this measure to be concentrated on a mass shell, and the energy turns out to be a function of the momentum. What the section cannot determine is the structure of $\mathcal{H}$ above each point of the mass shell, and in particular the spin. This is the subject of Section 6.4.

6.3.1 The Setting

The physical content of the hypothesis is the principle of relativity at the level of pure states. If two inertial observers describe the same system, the transition probabilities between its pure states, and more generally the whole Kähler geometry of $\mathbb{P}\mathcal{H}$, must be the same for both. The passage from one observer to the other acts therefore on $\mathbb{P}\mathcal{H}$ by Kähler isomorphisms, and this is the content of the following definition.

Definition 6.3.1 (Poincaré action)**.** *Let $\mathbb{P}\mathcal{H}$ be as per Definition* **??**. *A* Poincaré action *on $\mathbb{P}\mathcal{H}$ is a group homomorphism Φ from $\mathcal{P}$, as per Definition 6.1.7, to the group of Kähler isomorphisms of $\mathbb{P}\mathcal{H}$, written $\kappa \mapsto \Phi_\kappa$, such that every orbit map $\kappa \mapsto \Phi_\kappa(\phi)$ is continuous for the manifold topology of Proposition* **??**.

An elementary system cannot be separated into two independent sectors by the symmetry group. This is expressed in the same geometric terms as in the Galilean case.

Definition 6.3.2 (Irreducible Poincaré action)**.** *A Poincaré action Φ, as per Definition 6.3.1, is* irreducible *if the only closed complex subspaces $\mathcal{K} \subseteq \mathcal{H}$ such that $\Phi_\kappa(\mathbb{P}\mathcal{K}) \subseteq \mathbb{P}\mathcal{K}$ for all $\kappa \in \mathcal{P}$ are $\{0\}$ and $\mathcal{H}$.*

The two definitions fix the only structure on which the chapter relies. Since the confusion between what is given and what is derived is the main risk in this chapter, the following remark records it explicitly.

Remark 6.3.3 (Assumptions and outputs). The data of the chapter are the following.

- A separable Hilbert space $\mathcal{H}$, which enters only through the Kähler manifold $\mathbb{P}\mathcal{H}$ of Chapter **??**.
- An irreducible Poincaré action Φ on $\mathbb{P}\mathcal{H}$.
- Positivity of the energy, which is formulated in Definition 6.3.8 as soon as the generator of the time translations is available.

No operator on $\mathcal{H}$ is distinguished in advance. The four-momentum, the mass, the energy, the realization of $\mathcal{H}$ as a space of functions of the momentum and the spin are outputs, obtained in this section and in Section 6.4. Space inversion is not assumed on the state space. It enters in Section 6.5 as a requirement on a local realization of the fields.

The most delicate output is the spin. The next remark shows that it cannot be read from the state space, and it explains why the Hilbert space must be built from the action and not taken as given together with its realization.

Remark 6.3.4 (The projective space does not carry the spin). Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be separable infinite-dimensional Hilbert spaces. They are unitarily equivalent, since an orthonormal basis of the first is carried onto an orthonormal basis of the second [**TODO-Mor18**], and a unitary operator $W : \mathcal{H}_1 \to \mathcal{H}_2$ induces the map $[x] \mapsto [Wx]$, which preserves the inner product. By Equation (**??**) it preserves the metric, the symplectic form and the complex structure. It descends that $\mathbb{P}\mathcal{H}_1$ and $\mathbb{P}\mathcal{H}_2$ are isomorphic Kähler manifolds. The space $L^2(\mathcal{O}_m^+, \mu; \mathbb{C}^{2j+1})$ of Section 6.4 is separable and infinite-dimensional for every j, hence the same Kähler manifold hosts the states of particles of any spin. The value of j cannot be read from $\mathbb{P}\mathcal{H}$. It is a property of the action Φ, and Theorem 6.4 shows that it is the label of a representation of the stabilizer of the rest momentum.

6.3.2 The Lift and the Four-Momentum

The Wigner type argument that turns a Kähler action into a unitary ray representation is the same as in the Galilean case. What changes is the last step, the passage from ray representations to genuine ones, which in Chapter **??** required a central extension and here does not, by Corollary 6.2.4.

Proposition 6.3.5 (Unitary lift of a Poincaré action)**.** *Let Φ be a Poincaré action on $\mathbb{P}\mathcal{H}$. There exists a unique strongly continuous unitary representation $\Pi : \mathcal{P} \to U(\mathcal{H})$ such that $\Phi_\kappa([x]) = [\Pi(\kappa)x]$ for all $\kappa \in \mathcal{P}$ and $[x] \in \mathbb{P}\mathcal{H}$.*

Proof. As in the proof of Proposition **??**, each Φ_κ is induced by a unitary operator V_κ, unique up to a phase, by Proposition **??**, antiunitary operators being excluded because they reverse the sign of the symplectic form. The map $\kappa \mapsto [V_\kappa]$ is a homomorphism into $U(\mathcal{H})/U(1)$, and it is continuous in the topology of Theorem **??** because $|(x, V_\kappa y)|^2/(\|x\|^2\|y\|^2)$ is a transition probability, continuous in κ by the orbit-map hypothesis. Corollary 6.2.4 yields a strongly continuous unitary representation Π with $\Pi(\kappa)$ in the ray V_κ.

If Π' is another lift, then $\Pi' = \chi\,\Pi$ for a continuous homomorphism $\chi : \mathcal{P} \to U(1)$, which is smooth. Its differential $d\chi : \mathfrak{p} \to i\mathbb{R}$ is a Lie algebra homomorphism into an abelian algebra, hence it vanishes on $[\mathfrak{p},\mathfrak{p}] = \mathfrak{p}$ by item 2 of Corollary 6.1.16. Since $\mathcal{P}$ is connected, $\chi = 1$. □

Remark 6.3.6. Proposition 6.3.5 should be compared with Proposition **??**. There the lift carried a real parameter m, the class of a multiplier, and it was unique only up to $e^{i\epsilon s}$, which shifted the Hamiltonian by a constant. Here neither phenomenon occurs, because the Poincaré algebra has trivial second cohomology and no nonzero character, *i.e.*, it is perfect. In particular the energy operator is determined without additive ambiguity. The zero of the energy is fixed by the group, and the rest energy of a particle will turn out to be its mass. This is the first place where the cohomological computation of Section 6.2 pays off.

From now on Π denotes the lift of a fixed Poincaré action. The restriction of Π to the translations is a strongly continuous unitary representation of the abelian group $\mathbb{R}^4$, so that Theorem **??** applies in the same way as in Definition **??**. The joint spectrum of the translation generators is what a physicist calls the set of possible values of the four-momentum, and this is the object that the rest of the section determines.

Definition 6.3.7 (Four-momentum and positive energy)**.** *Let $T(a) := \Pi((a,\mathbb{1}))$ for $a \in \mathbb{R}^{1,3}$. By Theorem* **??** *there is a unique projection valued measure E on $\mathbb{R}^{1,3}$ such that*

$$T(a) = \int e^{ia\cdot p/\hbar}\, dE(p).$$

The selfadjoint operators $H := \int p^0\, dE$, $P_k := \int p^k\, dE$ and $P^2 := \int p\cdot p\, dE$ are the energy, the momentum and the squared four-momentum. The lift Π has positive energy *if H is bounded below.*

For $a = (s,\mathbf{0})$ and $a = (0,\mathbf{a})$ one recovers $T = e^{isH/\hbar}$ and $T = e^{-i\mathbf{a}\cdot\mathbf{P}/\hbar}$, in agreement with the active convention of Remark **??**. The hypotheses of the chapter can now be stated in full.

Definition 6.3.8 (Elementary relativistic system)**.** *An* elementary relativistic system *is an irreducible Poincaré action Φ, as per Definition 6.3.2, whose lift Π of Proposition 6.3.5 has positive energy, as per Definition 6.3.7.*

Positivity of the energy is the relativistic counterpart of the choice of the sign of m in item 3 of Theorem **??**. It plays the same role here, since it selects a sheet of the mass hyperboloid.

6.3.3 Covariance and the Mass Shell

The spectral measure E is built from the translations alone, whereas the Lorentz group acts on it by conjugation. The next lemma records how, and it is the link between the group structure and the geometry of the momenta.

Lemma 6.3.9 (Covariance of the spectral measure)**.** *For every Borel set $d \subseteq \mathbb{R}^{1,3}$ and every $g \in SL(2,\mathbb{C})$,*

$$\Pi((0,g))\,E(d)\,\Pi((0,g))^{-1} = E(\Lambda(g)d).$$

Consequently, for every Borel function $f : \mathbb{R}^{1,3} \to \mathbb{R}$, $\Pi((0,g))\,f(P)\,\Pi((0,g))^{-1} = \int f(\Lambda(g)^{-1}p)\,dE(p)$, as operators with their domains.

Proof. By item 2 of Lemma 6.1.10, $(0,g)(a,\mathbb{1})(0,g)^{-1} = (\Lambda(g)a,\mathbb{1})$, so that $\Pi((0,g))T(a)\Pi((0,g))^{-1} = T(\Lambda(g)a)$. The right-hand side equals $\int e^{i\Lambda(g)a\cdot p/\hbar}dE(p) = \int e^{ia\cdot q/\hbar}dE'(q)$ with $q = \Lambda(g)^{-1}p$ and $E'(d) := E(\Lambda(g)d)$. Hence E' and $\Pi((0,g))E(\cdot)\Pi((0,g))^{-1}$ are two projection valued measures with the same Fourier transform, and they coincide by the uniqueness in Theorem **??**. The second statement is the change of variables in the functional calculus, as in Lemma **??**. □

The lemma says that the support of E is a union of orbits of the Lorentz group in $\mathbb{R}^{1,3}$. The orbits relevant for a massive particle are the sheets of the mass hyperboloid, which we name together with their base point, the momentum of a particle at rest.

Definition 6.3.10 (Mass shell and rest momentum)**.** *For $m > 0$ the* forward mass shell *is $\mathcal{O}_m^+ := \{p \in \mathbb{R}^{1,3} \mid p^2 = m^2,\ p^0 > 0\}$, and $\mathcal{O}_m^- := -\mathcal{O}_m^+$. The* rest momentum *is $\mathring{p} := (m, \mathbf{0}) \in \mathcal{O}_m^+$. For $\mathbf{p} \in \mathbb{R}^3$ we set $\omega(\mathbf{p}) := \sqrt{|\mathbf{p}|^2 + m^2}$, so that $\mathbf{p} \mapsto (\omega(\mathbf{p}), \mathbf{p})$ is a homeomorphism $\mathbb{R}^3 \to \mathcal{O}_m^+$.*

To exploit the covariance we need to know that these sets are single orbits. The statement is elementary in the matrix model of Definition 6.1.1, since a Lorentz transformation acts there by $\hat{x} \mapsto g\hat{x}g^\dagger$, and its proof also covers the case of negative squared mass, which has to be excluded by positivity of the energy.

Lemma 6.3.11 (Orbits of the Lorentz group on the hyperboloids)**.** *Let $m > 0$ and $\nu > 0$.*

1. *$\Lambda(g)\mathcal{O}_m^\pm = \mathcal{O}_m^\pm$ for every $g \in SL(2,\mathbb{C})$, and $\Lambda(SL(2,\mathbb{C}))$ acts transitively on each of $\mathcal{O}_m^+$ and $\mathcal{O}_m^-$.*
2. *The hyperboloid $\{p \in \mathbb{R}^{1,3} \mid p^2 = -\nu^2\}$ is a single orbit of $\Lambda(SL(2,\mathbb{C}))$.*

Proof. Item 1. Let $p \in \mathcal{O}_m^+$. By item 3 of Lemma 6.1.2, $\hat{p}$ is positive definite, hence so is $g\hat{p}g^\dagger$, and its determinant is m^2. Therefore $\Lambda(g)p$ has square m^2 and time component $\frac{1}{2}\operatorname{tr}(g\hat{p}g^\dagger) > 0$, *i.e.*, it lies in $\mathcal{O}_m^+$. For transitivity, $\hat{p}/m$ is positive definite with determinant one. Let A be its positive definite square root, which has determinant one and belongs to $SL(2,\mathbb{C})$. Then $\widehat{\Lambda(A)\mathring{p}} = A(m\mathbb{1})A = \hat{p}$, so that $p = \Lambda(A)\mathring{p}$. Every point of $\mathcal{O}_m^+$ is thus in the orbit of $\mathring{p}$. The statement for $\mathcal{O}_m^-$ follows by applying the linear maps $\Lambda(g)$ to $-\mathcal{O}_m^+$.

Item 2. Let $p^2 = -\nu^2$. The matrix $\hat{p}$ is Hermitian with determinant $-\nu^2 < 0$, so its eigenvalues satisfy $\lambda_+ > 0 > \lambda_-$ and $\lambda_+\lambda_- = -\nu^2$. By the spectral theorem $\hat{p} = V \operatorname{diag}(\lambda_+, \lambda_-)V^\dagger$ with V unitary, and multiplying V by a phase we may take $\det V = 1$. The matrix $D := \operatorname{diag}\big(\sqrt{\lambda_+/\nu}, \sqrt{\nu/\lambda_+}\big)$ belongs to $SL(2,\mathbb{C})$ and, since $\lambda_- = -\nu^2/\lambda_+$, it satisfies $D\operatorname{diag}(\nu, -\nu)D = \operatorname{diag}(\lambda_+, \lambda_-)$. Hence $\hat{p} = g(\nu\sigma_3)g^\dagger$ with $g = VD$, and $\nu\sigma_3$ is the matrix of the vector $q = (0,0,0,\nu)$. It descends that $p = \Lambda(g)q$. □

The next proposition is the point at which the mass enters the theory, as an eigenvalue of a Casimir operator, *i.e.*, of an operator commuting with the whole representation. The mechanism is simple. The squared four-momentum is invariant under the Lorentz group and constant along the translations, so that P^2 commutes with everything, and by Schur's lemma it is a number.

Proposition 6.3.12 (The spectral measure lives on a mass shell)**.** *Let Φ be an elementary relativistic system, with lift Π and spectral measure E. Then*

1. *there is $m^2 \in \mathbb{R}$ with $P^2 = m^2\mathbb{1}$, and E is concentrated on $\{p \in \mathbb{R}^{1,3} \mid p^2 = m^2\}$,*
2. *$m^2 \geq 0$,*
3. *if $m > 0$, then $E(\mathbb{R}^{1,3} \setminus \mathcal{O}_m^+) = 0$ and $\operatorname{supp} E = \mathcal{O}_m^+$.*

Proof. Item 1. The function $p \mapsto p{\cdot}p$ is Lorentz invariant, so Lemma 6.3.9 yields $\Pi((0,g))e^{itP^2}\Pi((0,g))^{-1} = \int e^{it(\Lambda(g)^{-1}p)^2}dE(p) = e^{itP^2}$. The operator e^{itP^2} also commutes with every $T(a)$, since both are functions of E. As $(a,g) = (a,\mathbb{1})(0,g)$, e^{itP^2} commutes with $\Pi(\mathcal{P})$. The set $\Pi(\mathcal{P})$ is irreducible, since a closed invariant subspace $\mathcal{K}$ satisfies $\Phi_\kappa(\mathbb{P}\mathcal{K}) \subseteq \mathbb{P}\mathcal{K}$, and Definition 6.3.2 forces $\mathcal{K} \in \{\{0\}, \mathcal{H}\}$, as in Step 5 of the proof of Theorem **??**. Lemma **??** applied to $A = P^2$ yields $P^2 = m^2\mathbb{1}$ for some $m^2 \in \mathbb{R}$. Then $\int (p^2 - m^2)^2\, d\langle\psi, E\psi\rangle = \|(P^2 - m^2)\psi\|^2 = 0$ for every ψ in the domain, so $E(\{p^2 \neq m^2\}) = 0$.

Items 2 and 3. Let $S := \operatorname{supp} E$, as per Definition **??**. It is closed, nonempty because $E(\mathbb{R}^{1,3}) = \mathbb{1}$, contained in $\{p^2 = m^2\}$ by item 1, and invariant under $\Lambda(SL(2,\mathbb{C}))$ by Lemma 6.3.9. Suppose first that $m^2 < 0$. By item 2 of Lemma 6.3.11 the hyperboloid $\{p^2 = m^2\}$ is a single orbit, hence S is the whole hyperboloid, on which p^0 takes every real value. Suppose now that $m^2 > 0$ and that S meets $\mathcal{O}_m^-$. By item 1 of Lemma 6.3.11 the set $\mathcal{O}_m^-$ is a single orbit, so $\mathcal{O}_m^- \subseteq S$, and $p^0 \leq -m$ is unbounded below on it. In both cases the open sets $\{p^0 < -n\}$ meet S, hence $E(\{p^0 < -n\}) \neq 0$ for every $n \in \mathbb{N}$, and $H = \int p^0 dE$ is not bounded below. This contradicts positivity of the energy. Hence $m^2 \geq 0$, and for $m > 0$ one has $S \subseteq \mathcal{O}_m^+$. Since S is nonempty and invariant, and $\mathcal{O}_m^+$ is a single orbit, $S = \mathcal{O}_m^+$. □

Remark 6.3.13. The case $m = 0$ is excluded from now on. The null cone contains the origin and several orbits, and the stabilizer of a null vector is not $SU(2)$ but the covering of the Euclidean group of the plane, so that the construction of this chapter does not apply. The same exclusion was made for the Galilean mass in the case $m = 0$ of Remark **??**. The case $m^2 < 0$ is excluded by positivity of the energy, and it would describe tachyons.

Remark 6.3.14 (The mass as a label). Proposition 6.3.12 identifies the squared four-momentum as a Casimir operator and the mass as the label of an irreducible representation. This replaces the Galilean statement of Chapter **??**, where the mass labelled a cohomology class. A second Casimir, built from the Pauli–Lubanski vector, would distinguish the spins, but we do not need it. For $m > 0$ the spin is read from the representation of the stabilizer $SU(2)$ in Section 6.4.

6.3.4 The Hamiltonian of the Relativistic Free Particle

For $m > 0$ the Hamiltonian follows at once. The proof is much shorter than that of Theorem **??**, because the relation between boosts and time translations, which in the Galilean case was the engine of the proof, is already contained in the covariance of the spectral measure, and the rest is functional calculus.

Theorem 6.3 (Hamiltonian of the relativistic free particle)**.** *Let Φ be an elementary relativistic system, and let $m > 0$ be as in Proposition 6.3.12. With $\omega(\mathbf{p}) = \sqrt{|\mathbf{p}|^2 + m^2}$,*

$$H = \omega(\mathbf{P}) = \sqrt{|\mathbf{P}|^2 + m^2}, \qquad D(H) = D(|\mathbf{P}|). \tag{6.13}$$

Moreover $\sigma(H) = [m, \infty)$, and the spectral measure of $\mathbf{P}$ has support $\mathbb{R}^3$.

Proof. By Proposition 6.3.12, E is concentrated on $\mathcal{O}_m^+$ with support $\mathcal{O}_m^+$. The image of E under the homeomorphism $p \mapsto \mathbf{p}$ of $\mathcal{O}_m^+$ onto $\mathbb{R}^3$ is the spectral measure of $\mathbf{P}$, and its support is $\mathbb{R}^3$. Since $p^0 = \omega(\mathbf{p})$ on $\mathcal{O}_m^+$, $H = \int p^0 dE = \int \omega(\mathbf{p})\, dE = \omega(\mathbf{P})$. The domains coincide because $\omega(\mathbf{p})^2 = |\mathbf{p}|^2 + m^2$. Item 5 of Proposition **??** then yields $\sigma(H) = \overline{\omega(\mathbb{R}^3)} = [m, \infty)$. □

Remark 6.3.15 (Comparison with the Galilean case). Theorem 6.3 has the structure of Theorem **??**, with two differences. The energy has the absolute spectrum $[m, \infty)$ instead of $[E_0, \infty)$ with E_0 free, in accordance with Remark 6.3.6, and the mass is not a cocycle parameter but a Casimir eigenvalue, as announced in Remark **??**. Notice also that no boost relation was used explicitly. The statement that the energy is a function of the momentum alone is a consequence of the support of E being a single orbit, *i.e.*, of the transitivity of the Lorentz group on the mass shell.

The Nonrelativistic Limit

It remains to see that the Galilean theorem is recovered when the speed of light is sent to infinity. We restore c, so that the mass shell is $E^2 = c^2|\mathbf{p}|^2 + m^2c^4$ and the energy of Theorem 6.3 reads $H_c = \sqrt{c^2|\mathbf{P}|^2 + m^2c^4}$. The limit is a statement on the functional calculus of $\mathbf{P}$, and it identifies the constant E_0 which Galilean invariance left undetermined.

Proposition 6.3.16 (Nonrelativistic limit)**.** *In the hypotheses of Theorem 6.3, with c restored, one has $H_c - mc^2 \geq 0$, and for every $\psi \in D(|\mathbf{P}|^2)$*

$$\lim_{c\to\infty} \big(H_c - mc^2\big)\psi = \frac{|\mathbf{P}|^2}{2m}\,\psi.$$

*Hence the Galilean Hamiltonian of Theorem **??** is obtained with $E_0 = mc^2$, the rest energy.*

Proof. Put $h_c(\mathbf{p}) := \sqrt{c^2|\mathbf{p}|^2 + m^2c^4} - mc^2 = mc^2\big(\sqrt{1+x} - 1\big)$ with $x = |\mathbf{p}|^2/m^2c^2$. Since $0 \leq \sqrt{1+x} - 1 \leq x/2$ for $x \geq 0$, one has $0 \leq h_c(\mathbf{p}) \leq |\mathbf{p}|^2/2m$ for every c, and $h_c(\mathbf{p}) \to |\mathbf{p}|^2/2m$ pointwise as $c \to \infty$. Hence $H_c - mc^2 = h_c(\mathbf{P}) \geq 0$ and, by the functional calculus,

$$\Big\| \Big(h_c(\mathbf{P}) - \frac{|\mathbf{P}|^2}{2m}\Big)\psi \Big\|^2 = \int \Big(h_c(\mathbf{p}) - \frac{|\mathbf{p}|^2}{2m}\Big)^2 d\langle \psi, E^{(3)}\psi\rangle(\mathbf{p}),$$

where $E^{(3)}$ is the spectral measure of $\mathbf{P}$. The integrand tends to zero pointwise and it is bounded by $(|\mathbf{p}|^2/2m)^2$, which is integrable because $\psi \in D(|\mathbf{P}|^2)$. The claim follows by dominated convergence. □

The constant E_0 that Theorem **??** left free, and that Proposition **??** showed to be changeable at will in a Galilean action, has a meaning once the Galilei group is seen as the limit of the Poincaré group. It is the rest energy mc^2, which the Poincaré group fixes. In the Galilean world it has no dynamical effect, because it shifts H by a constant, and it is therefore invisible in the flow on $\mathbb{P}\mathcal{H}$. This is a statement on the Hamiltonians and not on the representations. That the representation theory of the Galilei group is the contraction of that of the Poincaré group is a more delicate statement, which we do not need and do not prove.

What Remains to Be Determined

Remark 6.3.17. Proposition 6.3.12 and Theorem 6.3 fix the joint spectrum of the translations, which is the mass shell, and the energy as a function of the momentum. They do not say how many independent states sit above each point of $\mathcal{O}_m^+$, *i.e.*, they do not fix the multiplicity of the spectral measure E, nor the way in which the Lorentz group acts on it. In the Galilean case this information was supplied by the von Neumann uniqueness theorem, which produced a tensor factor on which the rotations act irreducibly, as in Corollary **??**. Here the same role is played by the stabilizer of the rest momentum. Section 6.4 shows that irreducibility forces its action on the multiplicity space to be an irreducible representation of $SU(2)$, and this is where the spin emerges.

6.4 The Mass Shell, the Stabilizer and the Emergence of Spin

Section 6.3 determined the joint spectrum of the translations and the energy, but it left open the structure of $\mathcal{H}$ above each point of the mass shell. Two representations with the same spectral measure E can differ in the multiplicity of E and in the way the Lorentz group acts on it, and the Kähler manifold $\mathbb{P}\mathcal{H}$ cannot distinguish them by Remark 6.3.4. The missing information is contained in the action of the stabilizer of a point of the mass shell, which is the rotation group, on the fibre above that point. This is the mechanism of induced representations, and it is the relativistic counterpart of the tensor structure $L^2(\mathbb{R}^3) \otimes \mathcal{K}$ that Corollary **??** extracted from the Weyl relations.

The section proceeds in three steps. We first show that the mass shell is the homogeneous space of the Lorentz group with the rotation group as stabilizer, and we fix a global section given by the standard boosts. We then construct, for every pair (m, j), an induced representation on a space of functions of the momentum with values in a spin space. The failure of the section to be a homomorphism appears there as a cocycle of rotations, the Wigner rotation. The last step proves that every elementary relativistic system is one of these representations. The spin is therefore not postulated. It is read off as the label of the irreducible representation of the stabilizer which irreducibility forces, and the concrete realization of $\mathcal{H}$ is an output.

6.4.1 The Mass Shell as a Homogeneous Space

A massive particle at rest has four-momentum $\mathring{p}$, and any other massive particle of the same mass is obtained from it by a Lorentz transformation, as Lemma 6.3.11 shows. The stabilizer of $\mathring{p}$, *i.e.*, the group of Lorentz transformations leaving a particle at rest unchanged, is the group of rotations, and it will describe the internal degrees of freedom. This is the relativistic replacement of the picture of Chapter **??**, where the momenta also formed an orbit with the same stabilizer. The Galilean orbit is the flat space $\mathbb{R}^3$, boosts act on it by translations $\mathbf{p} \mapsto \mathbf{p} + m\mathbf{v}$ by Lemma **??**, and they form an abelian subgroup acting simply transitively. Here the orbit is a curved hyperboloid, and boosts do not form a subgroup, because the product of two boosts contains a rotation by Example 6.1.12. This is the geometric root of the Wigner rotation.

The following proposition singles out for each point p a preferred boost A_p which brings the rest momentum to p, the standard boost, and it shows that the rest of the group acts on p only through a rotation.

Proposition 6.4.1 (Stabilizer, standard boosts and orbit)**.** *Let $m > 0$.*

1. *$\Lambda(SL(2,\mathbb{C}))$ acts transitively on $\mathcal{O}_m^+$, and the stabilizer of $\mathring{p}$ in $SL(2,\mathbb{C})$ is $SU(2)$.*

2. *For $p \in \mathcal{O}_m^+$ let $\mathbf{w}_p := \operatorname{arsinh}(|\mathbf{p}|/m)\,\mathbf{p}/|\mathbf{p}|$ be its rapidity vector, with $\mathbf{w}_{\mathring{p}} = \mathbf{0}$, and let $A_p := \exp\big(\tfrac{1}{2}\mathbf{w}_p \cdot \boldsymbol{\sigma}\big)$. Then $A_p \in SL(2,\mathbb{C})$ is Hermitian and positive definite, $p \mapsto A_p$ is smooth, $A_{\mathring{p}} = \mathbb{1}$, $\Lambda(A_p)\mathring{p} = p$, and*

$$A_p = \Big(\frac{\hat{p}}{m}\Big)^{1/2} = \frac{\hat{p} + m\mathbb{1}}{\sqrt{2m(p^0+m)}}, \qquad A_p^2 = \frac{\hat{p}}{m}, \qquad A_p^{-2} = \frac{\tilde{p}}{m}. \tag{6.14}$$

3. *Every $g \in SL(2,\mathbb{C})$ can be written in a unique way as $g = A_p u$ with $p \in \mathcal{O}_m^+$ and $u \in SU(2)$, namely $p = \Lambda(g)\mathring{p}$. The map $g \mapsto (p,u)$ is a diffeomorphism $SL(2,\mathbb{C}) \to \mathcal{O}_m^+ \times SU(2)$. Consequently $gSU(2) \mapsto \Lambda(g)\mathring{p}$ is a diffeomorphism*

$$\mathcal{O}_m^+ \cong SL(2,\mathbb{C})/SU(2).$$

Proof. Item 1. Transitivity is item 1 of Lemma 6.3.11. Moreover $\Lambda(g)\mathring{p} = \mathring{p}$ if and only if $m\,g g^\dagger = m\,\mathbb{1}$, that is $g g^\dagger = \mathbb{1}$, which together with $\det g = 1$ means $g \in SU(2)$.

Item 2. Put $s := |\mathbf{p}|/m$, so that $\sinh|\mathbf{w}_p| = s$ and $\cosh|\mathbf{w}_p| = \sqrt{1+s^2} = p^0/m$. Hence $\hat{p}/m = \cosh|\mathbf{w}_p|\,\mathbb{1} + \sinh|\mathbf{w}_p|\,\frac{\mathbf{w}_p}{|\mathbf{w}_p|}\cdot\boldsymbol{\sigma} = \exp\big(\mathbf{w}_p\cdot\boldsymbol{\sigma}\big)$, and the positive square root of this matrix is A_p. The matrix A_p is the exponential of a traceless Hermitian matrix, so it is Hermitian, positive definite and has determinant one, and it is smooth in p as the positive square root of $\hat{p}/m$, which is positive definite by item 3 of Lemma 6.1.2. The Cayley–Hamilton identity $\hat{p}^2 = 2p^0\hat{p} - m^2\mathbb{1}$ yields $(\hat{p}+m\mathbb{1})^2 = 2(p^0+m)\hat{p}$, which is the explicit formula. Moreover $\widehat{\Lambda(A_p)\mathring{p}} = A_p(m\mathbb{1})A_p = mA_p^2 = \hat{p}$, and $A_p^{-2} = m\hat{p}^{-1} = \tilde{p}/m$ by Lemma 6.1.2.

Item 3. By Lemma 6.1.3, every g is uniquely $g = \exp\big(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\big)u$. Let $p := m(\cosh|\mathbf{w}|,\ \sinh|\mathbf{w}|\,\mathbf{w}/|\mathbf{w}|) \in \mathcal{O}_m^+$, so that $\mathbf{w} = \mathbf{w}_p$. Then $\hat{p} = m\,e^{\mathbf{w}\cdot\boldsymbol{\sigma}} = m\,gg^\dagger$, hence $p = \Lambda(g)\mathring{p}$ and $\exp\big(\tfrac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\big) = A_p$. The map $\mathbf{w} \mapsto p$ is a diffeomorphism of $\mathbb{R}^3$ onto $\mathcal{O}_m^+$, with inverse $p \mapsto \mathbf{w}_p$, so $g \mapsto (p,u)$ is the composition of the diffeomorphism of Lemma 6.1.3 with a diffeomorphism. The map $g \mapsto \Lambda(g)\mathring{p}$ is constant on the cosets $gSU(2)$ and has the smooth inverse $p \mapsto A_p SU(2)$. □

Example 6.4.2 (The standard boost along an axis). Let $p = m(\cosh w, 0, 0, \sinh w)$. Then $\mathbf{w}_p = w\,\mathbf{e}_3$, $A_p = \operatorname{diag}(e^{w/2}, e^{-w/2})$, $\hat{p} = m\operatorname{diag}(e^w, e^{-w})$ and $\tilde{p} = m\operatorname{diag}(e^{-w}, e^w)$, so that $A_p^2 = \hat{p}/m$ and $A_p^{-2} = \tilde{p}/m$ as in Equation (6.14). The rapidity vector $\mathbf{w}_p$ is the Minkowskian analogue of the velocity of the Galilean boost of Chapter **??**, since it measures how far, in the hyperbolic sense, the particle is from rest.

Proposition 6.4.1 says that $SL(2,\mathbb{C})$ fibres over the mass shell with fibre $SU(2)$, and that $p \mapsto A_p$ is a global section. This section is not a homomorphism, and the failure of this property produces the Wigner rotation. That the section is global, and the fibration trivial, allows us to describe the induced representation by functions on the mass shell with values in a fixed vector space. The last ingredient is the measure on the orbit, which is needed to define the Hilbert space.

Lemma 6.4.3 (Invariant measure)**.** *The measure $d\mu(p) := d^3\mathbf{p}\,/\,2\omega(\mathbf{p})$ on $\mathcal{O}_m^+ \cong \mathbb{R}^3$ is invariant under $\Lambda(g)$ for every $g \in SL(2,\mathbb{C})$.*

Proof. The Lebesgue measure d^4p on $\mathbb{R}^{1,3}$ is invariant under $\Lambda(g)$, since its determinant is 1, and p^2 and the sign of p^0 are invariant as well. Hence the measure $\delta(p^2 - m^2)\,d^4p$, restricted to $p^0 > 0$, is invariant. Integrating over p^0 with $\delta((p^0)^2 - \omega(\mathbf{p})^2) = \delta(p^0 - \omega(\mathbf{p}))/2\omega(\mathbf{p})$ yields $d\mu$. □

The factor $1/2\omega$ is the price of using the three-momentum as a coordinate on a curved orbit. The flat measure $d^3\mathbf{p}$ is not Lorentz invariant, and the correction is exactly the one that appears in Section 6.7 when the Euclidean and the covariant norms are compared.

6.4.2 Candidates: Induced Representations

A representation with the spectral measure of Section 6.3.3 is determined by what the stabilizer does on the fibre above the rest momentum. We construct all the candidates explicitly. By Proposition 6.4.1, $SL(2,\mathbb{C}) \to \mathcal{O}_m^+$ is a principal $SU(2)$-bundle with the global section $p \mapsto A_p$. A state with spin is a section of the vector bundle associated to it by a representation of $SU(2)$, and such a section is described by a function on the whole group $SL(2,\mathbb{C})$ with values in a spin space, which is rigidly transported along the cosets of $SU(2)$. We restrict it afterwards to the standard boosts. The geometric formulation is developed in

Appendix 7, and here we record only the formulas which are used in the chapter. The spin enters through a parameter j which labels the candidates. Theorem 6.4 will show that this parameter is not an input, because each elementary system realizes exactly one value.

Definition 6.4.4 (Spin data)**.** *Let $j \in \{0, \frac{1}{2}, 1, \dots\}$. We set $V := \mathbb{C}^{2j+1}$ and we denote by $\pi := d_j$ the irreducible unitary representation of $SU(2)$ of dimension $2j+1$ of Corollary* **??**. *For $j = \frac{1}{2}$ we have $V = \mathbb{C}^2$ and $\pi(u) = u$.*

Definition 6.4.5 (Equivariant functions)**.** *Let $\mathcal{E}_\pi$ be the space of Borel functions $f : SL(2,\mathbb{C}) \to V$ such that*

$$f(gu) = \pi(u^{-1}) f(g), \qquad g \in SL(2,\mathbb{C}),\ u \in SU(2). \tag{6.15}$$

For $g \in SL(2,\mathbb{C})$ the left translation *is $(\mathcal{L}(g)f)(g') := f(g^{-1}g')$.*

Left translations preserve $\mathcal{E}_\pi$, because they commute with the right multiplication by $SU(2)$, and $\mathcal{L}(g)\mathcal{L}(g') = \mathcal{L}(gg')$. The norm $\|f(g)\|$ is constant on each coset $gSU(2)$, since π is unitary, so that it is a function on $\mathcal{O}_m^+$. By Proposition 6.4.1, an equivariant function is determined by its restriction to the standard boosts.

Lemma 6.4.6 (Equivariant functions are sections)**.** *The map $f \mapsto \psi_f$, $\psi_f(p) := f(A_p)$, is a bijection from $\mathcal{E}_\pi$ onto the Borel functions $\mathcal{O}_m^+ \to V$, with inverse $\psi \mapsto f_\psi$, $f_\psi(A_p u) := \pi(u^{-1})\psi(p)$.*

Proof. By Proposition 6.4.1 every g is uniquely of the form $A_p u$, so f_ψ is well defined and Borel. It satisfies Equation (6.15), since $f_\psi(A_p uu') = \pi(u'^{-1})\pi(u^{-1})\psi(p) = \pi(u'^{-1}) f_\psi(A_p u)$. Conversely, Equation (6.15) yields $f(A_p u) = \pi(u^{-1}) f(A_p)$, so $f = f_{\psi_f}$. □

Definition 6.4.5 and Lemma 6.4.6 say that $\mathcal{E}_\pi$ is the space of sections of the associated bundle $SL(2,\mathbb{C}) \times_{SU(2)} V$ over $\mathcal{O}_m^+$, by Theorem 7.1, and that the standard boosts trivialize it, by Lemma 7.5.2. This identification explains the form of the transformation law which is computed next.

The action of the left translations on the sections is the central computation of the section. It produces the cocycle of rotations in which the failure of $p \mapsto A_p$ to be a homomorphism is recorded.

Proposition 6.4.7 (Induced action and Wigner rotation)**.** *For $g \in SL(2,\mathbb{C})$ and $p \in \mathcal{O}_m^+$ set*

$$R_W(g,p) := A_p^{-1}\, g\, A_{\Lambda(g)^{-1}p}. \tag{6.16}$$

Then $R_W(g,p) \in SU(2)$, and for all $f \in \mathcal{E}_\pi$

$$\psi_{\mathcal{L}(g)f}(p) = \pi\big(R_W(g,p)\big)\, \psi_f\big(\Lambda(g)^{-1}p\big). \tag{6.17}$$

Moreover $R_W(gg',p) = R_W(g,p) R_W(g', \Lambda(g)^{-1}p)$, $R_W(\mathbb{1},p) = \mathbb{1}$, and $R_W(u,p) = u$ for every $u \in SU(2)$.

Proof. Let $q := \Lambda(g)^{-1}p$. Since $\Lambda(g^{-1}A_p)\mathring{p} = \Lambda(g)^{-1}p = q$, Proposition 6.4.1 yields $g^{-1}A_p = A_q h$ with $h \in SU(2)$, namely $h = A_q^{-1} g^{-1} A_p = R_W(g,p)^{-1}$. Hence $R_W(g,p) \in SU(2)$ and

$$\psi_{\mathcal{L}(g)f}(p) = f(g^{-1}A_p) = f(A_q h) = \pi(h^{-1}) f(A_q) = \pi(R_W(g,p))\, \psi_f(q).$$

For the cocycle relation, with $q = \Lambda(g)^{-1}p$,

$$R_W(g,p) R_W(g',q) = A_p^{-1} g A_q A_q^{-1} g' A_{\Lambda(g')^{-1}q} = A_p^{-1} gg' A_{\Lambda(gg')^{-1}p} = R_W(gg',p).$$

The identity $R_W(\mathbb{1},p) = \mathbb{1}$ is immediate. If $u \in SU(2)$, then $u^{-1}A_p u$ is positive definite with square $u^{-1}\hat{p}u/m = \widehat{\Lambda(u)^{-1}p}/m$, so that $A_{\Lambda(u)^{-1}p} = u^{-1}A_p u$ and $R_W(u,p) = A_p^{-1} uu^{-1} A_p u = u$. □

Equation (6.17) shows that on the section ψ_f the translation by g acts by a rigid rotation of the value, by $R_W(g,p)$, composed with the relocation of the argument. For rotations $R_W(u,p) = u$ does not depend on p. In the language of Appendix 7, $R_W(g,p)^{-1}$ is the transition function of the trivialization $p \mapsto A_p$, by Proposition 7.6.4. For boosts it depends on p, and this dependence is studied in Proposition 6.5.1, because it is the origin of the lack of locality treated in Section 6.5. Here only the cocycle identity is needed, since it makes the following formula a representation.

Definition 6.4.8 (Induced representation)**.** *Let μ be the measure of Lemma 6.4.3 and $\mathcal{H}_{m,j} := L^2(\mathcal{O}_m^+, \mu; V)$. For $(a,g) \in \mathcal{P}$ and $\psi \in \mathcal{H}_{m,j}$ we set*

$$\big(U(a,g)\psi\big)(p) := e^{ia\cdot p/\hbar}\,\pi\big(R_W(g,p)\big)\,\psi\big(\Lambda(g)^{-1}p\big). \tag{6.18}$$

We write $U_{m,j}$ when the dependence on m and j has to be displayed.

Proposition 6.4.9. *U is a strongly continuous unitary representation of $\mathcal{P}$ on $\mathcal{H}_{m,j}$. Its spectral measure of the translations is $E(d) = \chi_d$, the multiplication by the characteristic function of $d \subseteq \mathcal{O}_m^+$, so that $P^2 = m^2\mathbb{1}$, H is the multiplication by $\omega(\mathbf{p})$, and the energy is positive.*

Proof. Homomorphism. The translations act by multiplication by $e^{ia\cdot p/\hbar}$. Using the invariance $a'\cdot\Lambda(g)^{-1}p = \Lambda(g)a'\cdot p$ of the product and the cocycle relation of Proposition 6.4.7,

$$\begin{aligned}\big(U(a,g)U(a',g')\psi\big)(p) &= e^{ia\cdot p/\hbar}\pi(R_W(g,p))\,e^{i\Lambda(g)a'\cdot p/\hbar}\pi\big(R_W(g',\Lambda(g)^{-1}p)\big)\psi\big(\Lambda(gg')^{-1}p\big)\\ &= e^{i(a+\Lambda(g)a')\cdot p/\hbar}\pi\big(R_W(gg',p)\big)\psi\big(\Lambda(gg')^{-1}p\big) = \big(U((a,g)(a',g'))\psi\big)(p).\end{aligned}$$

Unitarity. Since $\pi(R_W(g,p))$ is unitary and μ is invariant by Lemma 6.4.3,

$$\|U(a,g)\psi\|^2 = \int \|\psi(\Lambda(g)^{-1}p)\|^2\,d\mu(p) = \|\psi\|^2,$$

and $U(a,g)$ is invertible with inverse $U((a,g)^{-1})$.

Strong continuity. For ψ continuous with compact support, $(a,g,p) \mapsto e^{ia\cdot p/\hbar}\pi(R_W(g,p))\psi(\Lambda(g)^{-1}p)$ is continuous. For (a,g) in a compact neighbourhood its support lies in a fixed compact set, and it is bounded by $\sup\|\psi\|$. By dominated convergence $(a,g) \mapsto U(a,g)\psi$ is continuous in L^2. Such ψ are dense and the $U(a,g)$ are isometries, so continuity holds for every ψ.

The last statement is read off from Equation (6.18) with $g = \mathbb{1}$. □

The central element of Proposition 6.2.5 acts in the induced representation in a transparent way, and this is where the topological sign of Section 6.2 meets the spin.

Corollary 6.4.10 (The central sign is $(-1)^{2j}$)**.** *In the representation $U_{m,j}$, $U(0,-\mathbb{1}) = (-1)^{2j}\,\mathbb{1}$. Hence, if $U_{m,j}$ is irreducible, its central sign of Proposition 6.2.5 is $s_\Pi = +1$ for integer j, in which case it factors through the proper orthochronous Poincaré group $\bar{\mathcal{P}}_0$, and $s_\Pi = -1$ for half-integer j.*

Proof. Since $\Lambda(-\mathbb{1}) = \mathrm{id}$, Equation (6.18) yields $(U(0,-\mathbb{1})\psi)(p) = \pi(R_W(-\mathbb{1},p))\psi(p)$, and $R_W(-\mathbb{1},p) = A_p^{-1}(-\mathbb{1})A_p = -\mathbb{1}$. The representation d_j satisfies $d_j(-\mathbb{1}) = (-1)^{2j}\mathbb{1}$. □

6.4.3 Spin as a Label: the Structure Theorem

Up to this point j has been a parameter ranging over the candidates, and nothing guarantees that an elementary relativistic system is one of them. The next theorem closes this gap. It is the classification theorem of Wigner and Mackey, and it is the relativistic counterpart of Corollary **??**. Its content for the present chapter is that the Hilbert space above the mass shell and the spin are determined by the action Φ.

Theorem 6.4 (Wigner, Mackey)**.** *Let $m > 0$.*

1. *For every $j \in \{0, \tfrac{1}{2}, 1, \dots\}$ the representation $U_{m,j}$ of Definition 6.4.8 is an irreducible strongly continuous unitary representation of $\mathcal{P}$ of positive energy with $P^2 = m^2\mathbb{1}$.*

2. *Let Φ be an elementary relativistic system with lift Π and mass $m > 0$ as per Proposition 6.3.12. Then there exist a unique $j \in \{0, \tfrac{1}{2}, 1, \dots\}$ and a unitary operator $\mathcal{W} : \mathcal{H} \to \mathcal{H}_{m,j}$ such that $\mathcal{W}\Pi(\kappa)\mathcal{W}^{-1} = U_{m,j}(\kappa)$ for all $\kappa \in \mathcal{P}$ and $\mathcal{W}E(d)\mathcal{W}^{-1} = \chi_d$ for every Borel set d.*

See [**TODO-Wigner1939**] and [**TODO-Mackey1958**]. The proof is not reproduced here. The following remark shows where each hypothesis of the theorem enters.

Remark 6.4.11 (Reading the theorem in four steps). *Step 1, a system of imprimitivity.* By Lemma 6.3.9 and Proposition 6.3.12, E is a projection valued measure on $\mathcal{O}_m^+$ with $\Pi((0,g))E(d)\Pi((0,g))^{-1} = E(\Lambda(g)d)$. Hence the pair formed by the restriction of Π to $SL(2,\mathbb{C})$ and by E is a system of imprimitivity for $SL(2,\mathbb{C})$ based on the homogeneous space $\mathcal{O}_m^+ \cong SL(2,\mathbb{C})/SU(2)$ of Proposition 6.4.1.

Step 2, the induced form. Mackey's imprimitivity theorem identifies $\mathcal{H}$ with $L^2(\mathcal{O}_m^+, \mu; \mathcal{K})$, for a separable Hilbert space $\mathcal{K}$, *i.e.*, with the square integrable sections of a bundle of the type of Appendix 7, and $SL(2,\mathbb{C})$ acts as in Proposition 6.4.7, with a unitary representation d of $SU(2)$ on $\mathcal{K}$ in place of π. The measure class is the unique quasi-invariant class on the homogeneous space, and μ of Lemma 6.4.3 is invariant, so that no Radon–Nikodym factor appears.

Step 3, the translations. Since $T(a) = \int e^{ia\cdot p/\hbar} dE(p)$, the translations act by multiplication by $e^{ia\cdot p/\hbar}$, and Π is given by Equation (6.18) with $(d, \mathcal{K})$ in place of (π, V).

Step 4, irreducibility selects the spin. If $\mathcal{K}_1 \subseteq \mathcal{K}$ is a closed d-invariant subspace, then $L^2(\mathcal{O}_m^+, \mu; \mathcal{K}_1)$ is Π-invariant, so irreducibility of Π forces d to be irreducible. Since $SU(2)$ is compact, d is finite-dimensional and equivalent to d_j for a unique j [**TODO-Hal03**], so that $\mathcal{K} \cong \mathbb{C}^{2j+1}$. Equivalence classes of induced representations are in bijection with equivalence classes of the inducing representations, which yields the uniqueness of j. Conversely, d_j irreducible yields $U_{m,j}$ irreducible.

Remark 6.4.12 (Spin is an output). Theorem 6.4 answers the question left open by Remark 6.3.4. The state space $\mathbb{P}\mathcal{H}$ does not contain the spin, which is a property of the action Φ, and it enters as the label of the irreducible representation of the stabilizer of the rest momentum that irreducibility forces. The mechanism is the same as in Corollary **??**, where the multiplicity space of the Weyl representation carried a representation of $SU(2)$ which irreducibility forced to be d_j. The difference lies in the boosts. In the Galilean case $B(\mathbf{v})$ acts on $\mathbb{C}^{2j+1}$ trivially, whereas here boosts act on the fibre through the Wigner rotation $R_W(g,p)$. By Corollary 6.4.10, integer and half-integer spin are distinguished by the value $s_\Pi = (-1)^{2j}$ of the central sign of Proposition 6.2.5.

Remark 6.4.13. The theorem identifies an elementary relativistic system with a pair (m, j), exactly as in Chapter **??** an elementary Galilean system was labelled by (m, j) after Corollary **??**. Combined with Proposition 6.3.5, Proposition 6.3.12 and Theorem 6.3, it says that an elementary relativistic system of mass $m > 0$ is realized, up to unitary equivalence, on $\mathcal{H}_{m,j}$ with $H = \omega(\mathbf{P})$. From now on we take $j = \frac{1}{2}$, which is the case of the Dirac equation, so that $V = \mathbb{C}^2$ and $\pi(u) = u$.

6.5 Locality and the Spinorial Extension

Section 6.4 showed that an elementary relativistic system is, up to unitary equivalence, one of the representations $U_{m,j}$ of Definition 6.4.8, so that the Hilbert space above the mass shell and the spin are determined by the action. Such a representation is unitary and irreducible, and it contains all the content of the system. Its defect is that the transformation law (6.18) involves the matrix $\pi(R_W(g,p))$, which depends on the integration variable p once the state is written in position space. In this section we make this precise, we prove that the defect cannot be removed as long as the transformation matrices are required to be unitary, and we show that the minimal way out which respects space inversion is the Dirac space $\mathbb{C}^4$.

The section proceeds as follows. We first show that the Wigner rotation depends on the momentum, which is the precise form of the defect. We then define local covariant realizations and we record how space inversion acts on the Lorentz group, since the classification of the finite-dimensional representations that can serve as transformation matrices makes use of it. Space inversion enters at the end as a requirement on the realization, and it selects the Dirac space.

The Wigner Rotation Depends on the Momentum

By Theorem 6.4, the spin of an elementary system is carried by a representation of the stabilizer of the rest momentum. The stabilizer reaches the fibre above a generic momentum p only through the Wigner rotation $R_W(g,p)$ of Equation (6.16), and the question is whether this matrix is independent of p. For rotations it is, since $R_W(u,p) = u$ by Proposition 6.4.7. The next proposition shows that for boosts it is not, and this is the defect which the rest of the section removes.

Proposition 6.5.1 (Momentum dependence of the Wigner rotation)**.** *Let $q \in \mathcal{O}_m^+$ with $\mathbf{q} \neq \mathbf{0}$, and let $p' \in \mathcal{O}_m^+$ and $p := \Lambda(A_q)p'$. Then $R_W(A_q, p) = \mathbb{1}$ if and only if $\mathbf{q} \times \mathbf{p}' = \mathbf{0}$. In particular $p \mapsto R_W(A_q, p)$ is not constant, since $R_W(A_q, \mathring{p}) = \mathbb{1}$ whereas $R_W(A_q, p) \neq \mathbb{1}$ if $\mathbf{p}'$ is not parallel to $\mathbf{q}$.*

Proof. Since $\Lambda(A_q)^{-1}p = p'$, $R_W(A_q, p) = A_p^{-1}A_qA_{p'}$, which equals $\mathbb{1}$ if and only if $A_p = A_qA_{p'}$. If this holds, $A_qA_{p'}$ is Hermitian, hence $A_qA_{p'} = (A_qA_{p'})^\dagger = A_{p'}A_q$, so $\hat{q} = mA_q^2$ and $\hat{p}' = mA_{p'}^2$ commute. Using $[\mathbf{a}\cdot\boldsymbol{\sigma}, \mathbf{b}\cdot\boldsymbol{\sigma}] = 2i(\mathbf{a}\times\mathbf{b})\cdot\boldsymbol{\sigma}$ we obtain $\mathbf{q}\times\mathbf{p}' = \mathbf{0}$. Conversely, if $\mathbf{q}\times\mathbf{p}' = \mathbf{0}$, then $\hat{q}$ and $\hat{p}'$ commute, and so do A_q and $A_{p'}$, which are polynomials in them by Equation (6.14). Then $A_qA_{p'}$ is positive definite and Hermitian, and $\Lambda(A_qA_{p'})\mathring{p} = \Lambda(A_q)p' = p$, so $A_qA_{p'} = A_p$ by the uniqueness in item 3 of Proposition 6.4.1. For the last claim, $\mathring{p} = \Lambda(A_q)p'$ with $p' = \mathsf{P}q$, and $\mathbf{p}' = -\mathbf{q}$ is parallel to $\mathbf{q}$. □

Remark 6.5.2. Proposition 6.5.1 is the Thomas–Wigner rotation of Example 6.1.12. The composition of two boosts with non-collinear directions is a boost followed by a rotation. For $j = \frac{1}{2}$ the representation π is the defining one, so $\pi(R_W(A_q, \cdot))$ is not constant either. Thus a boost acts on the spin of a particle of momentum p by a rotation which depends on p, and it is not possible to speak of the spin of the particle independently of its momentum. In the Galilean case no such effect exists, because $B(\mathbf{v})$ acts on $\mathbb{C}^{2j+1}$ trivially.

Local Covariant Realizations

A transformation law is local if the value of the field at a point is mapped to the value at the transformed point by a matrix which does not depend on the point. In momentum space this means a matrix independent of p. We therefore look for realizations of the induced representation of the following type.

Definition 6.5.3 (Local covariant realization)**.** *Let (π, V) be as in Definition 6.4.4. A* local covariant realization *of (π, V) is a triple (D, W, ι), where W is a finite-dimensional complex vector space, $D : SL(2,\mathbb{C}) \to GL(W)$ is a continuous homomorphism, not required to be unitary, and $\iota : V \to W$ is an injective linear map with*

$$\iota\,\pi(u) = D(u)\,\iota, \qquad u \in SU(2). \tag{6.19}$$

We set $F(\mathring{p}) := \iota(V)$ and $F(p) := D(A_p)F(\mathring{p})$ for $p \in \mathcal{O}_m^+$, and we denote by $\mathcal{F}_D$ the space of Borel fields $\Psi : \mathcal{O}_m^+ \to W$ with $\Psi(p) \in F(p)$ for all p. The group acts on $\mathcal{F}_D$ by

$$\big(\mathcal{U}_D(a,g)\Psi\big)(p) := e^{ia\cdot p/\hbar}\,D(g)\,\Psi\big(\Lambda(g)^{-1}p\big). \tag{6.20}$$

The matrix $D(g)$ in Equation (6.20) does not depend on p, in contrast with $\pi(R_W(g,p))$ in Equation (6.18). The price is that the fields are constrained to lie in $F(p)$. The two descriptions are equivalent, and the equivalence is where Equation (6.19) is used.

Lemma 6.5.4 (Local realization of the induced representation)**.** *Let (D, W, ι) be a local covariant realization of (π, V). The map*

$$(J\psi)(p) := D(A_p)\,\iota\,\psi(p)$$

is a linear bijection from the Borel functions $\mathcal{O}_m^+ \to V$ onto $\mathcal{F}_D$, and, with $U(a,g)$ acting on Borel functions by Equation (6.18)*,*

$$J\,U(a,g) = \mathcal{U}_D(a,g)\,J. \tag{6.21}$$

In particular $\mathcal{F}_D$ is invariant under $\mathcal{U}_D$, and $\mathcal{U}_D$ is a representation of $\mathcal{P}$.

Proof. J is linear, with values in $\mathcal{F}_D$, and it is bijective because ι is injective and $D(A_p)$ is invertible: the inverse is $\psi(p) = \iota^{-1}D(A_p)^{-1}\Psi(p)$ on the range of ι. For Equation (6.21), let $q := \Lambda(g)^{-1}p$. By Equation (6.16), $gA_q = A_pR_W(g,p)$, and by Equation (6.19)

$$\begin{aligned}\big(\mathcal{U}_D(a,g)J\psi\big)(p) &= e^{ia\cdot p/\hbar}D(g)D(A_q)\,\iota\,\psi(q) = e^{ia\cdot p/\hbar}D(A_p)D\big(R_W(g,p)\big)\iota\,\psi(q)\\ &= e^{ia\cdot p/\hbar}D(A_p)\,\iota\,\pi\big(R_W(g,p)\big)\psi(q) = \big(J\,U(a,g)\psi\big)(p).\end{aligned}$$

Hence $\mathcal{U}_D(a,g)$ maps $\mathcal{F}_D$ into itself, and $\mathcal{U}_D = JUJ^{-1}$ is a representation. □

Remark 6.5.5 (Local realizations as subbundles). In the language of Appendix 7, Lemma 6.5.4 says that J is an isomorphism of the associated bundle $E_\pi = SL(2,\mathbb{C}) \times_{SU(2)} V$ onto the subbundle of the trivial bundle $\mathcal{O}_m^+ \times W$ with fibres $F(p)$. It intertwines the action of $SL(2,\mathbb{C})$ on the sections of E_π with the action $(g\cdot\Psi)(p) = D(g)\Psi(\Lambda(g)^{-1}p)$ on the sections of the trivial bundle. The matrix $D(g)$ does not depend on p, and the price is that the image is a subbundle whose fibres move with p, with $D(g)F(p) = F(\Lambda(g)p)$. The problem of this section is therefore the search for a representation D and an $SL(2,\mathbb{C})$-invariant subbundle F which is isomorphic to E_π.

Remark 6.5.6 (Locality in position space). For a field Ψ for which the integral converges, for instance continuous with compact support on $\mathcal{O}_m^+$, set $\Psi(x) := \int e^{-ip\cdot x/\hbar}\Psi(p)\,d\mu(p)$. The change of variables $p = \Lambda(g)q$ and the invariance of μ yield

$$\big(\mathcal{U}_D(a,g)\Psi\big)(x) = D(g)\,\Psi\big(\Lambda(g)^{-1}(x-a)\big).$$

This is a local transformation law: the value at x is transformed into the value at the transformed point by the fixed matrix $D(g)$. For the induced representation the same computation yields

$$\int e^{-ip\cdot x/\hbar}\,\pi\big(R_W(g,p)\big)\,\psi\big(\Lambda(g)^{-1}p\big)\,d\mu(p),$$

in which the matrix depends on the integration variable and cannot be taken out of the integral. By Proposition 6.5.1 it is not constant in p. A change of the section $p \mapsto A_p$ would replace R_W by a cohomologous cocycle and does not change this conclusion, see Proposition 6.5.10.

Space Inversion on the Lorentz Group

The two Weyl representations of the next subsection are exchanged by an automorphism of $SL(2,\mathbb{C})$ which comes from space inversion. Since this automorphism enters already in the classification of the finite-dimensional representations, we introduce it before them, together with its action on the mass shell. Space inversion plays no role in the Poincaré group $\mathcal{P}$, and its action on $SL(2,\mathbb{C})$ is determined by the matrix model. Inversion reverses the spatial part of a vector, so that the matrix $\hat{x}$ goes into $\tilde{x}$. On the group side this is the exchange of g with $(g^\dagger)^{-1}$, an automorphism which is not inner, *i.e.*, it does not come from conjugation by an element of $SL(2,\mathbb{C})$. This is what makes parity a nontrivial constraint.

Lemma 6.5.7 (Space inversion). *Let* $\mathsf{P}(x^0,\mathbf{x}) := (x^0,-\mathbf{x})$ *and* $\theta(g) := (g^\dagger)^{-1}$ *for* $g \in SL(2,\mathbb{C})$.

1. $\widehat{\mathsf{P}x} = \tilde{x}$. *The map* θ *is an automorphism of* $SL(2,\mathbb{C})$ *with* $\theta^2 = \mathrm{id}$ *and* $\theta|_{SU(2)} = \mathrm{id}$, *and*
$$\Lambda(\theta(g)) = \mathsf{P}\,\Lambda(g)\,\mathsf{P}. \tag{6.22}$$
2. $\mathsf{P}\mathcal{O}_m^+ = \mathcal{O}_m^+$, *and for* $p \in \mathcal{O}_m^+$
$$\theta(A_p) = A_{\mathsf{P}p} = A_p^{-1}. \tag{6.23}$$
3. *The measure* μ *of Lemma 6.4.3 is invariant under* P.

Proof. Item 1. The first claim is the definition of $\tilde{x}$. If $g \in SU(2)$ then $g^\dagger = g^{-1}$, so $\theta(g) = g$. By Equation (6.5), $\widehat{\mathsf{P}\Lambda(g)x} = \widehat{\Lambda(g)x} = (g^\dagger)^{-1}\tilde{x}g^{-1} = \theta(g)\,\widehat{\mathsf{P}x}\,\theta(g)^\dagger$, which is Equation (6.22) after replacing x by $\mathsf{P}x$.

Item 2. P preserves p^2 and p^0. Since A_p is Hermitian, $\theta(A_p) = A_p^{-1}$, which is positive definite. By Equation (6.22) and $\mathsf{P}\mathring{p} = \mathring{p}$, $\Lambda(\theta(A_p))\mathring{p} = \mathsf{P}\Lambda(A_p)\mathring{p} = \mathsf{P}p$. The uniqueness in item 3 of Proposition 6.4.1 yields $\theta(A_p) = A_{\mathsf{P}p}$.

Item 3. In the coordinate $\mathbf{p}$ the map P is $\mathbf{p} \mapsto -\mathbf{p}$, which preserves $d^3\mathbf{p}$ and $\omega(\mathbf{p})$. □

In words, space inversion exchanges the standard boost with its inverse. The boost that brings a particle from rest to momentum $\mathbf{p}$ and the one that brings it to $-\mathbf{p}$ are inverse to each other.

Finite-Dimensional Representations of the Lorentz Group

The next step is to find the finite-dimensional representations of $SL(2,\mathbb{C})$ that can serve as D. They are classified through the chiral decomposition of Lemma 6.1.15, which showed in Section 6.1 that the complexified Lorentz algebra is the sum of two commuting copies of $\mathfrak{su}(2)$: a representation is labelled by a pair of spins, one for each copy. The differential of a continuous finite-dimensional representation D is a homomorphism of real Lie algebras, which we extend complex-linearly to $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{C}}$, and the generators A_k and B_k are those of Lemma 6.1.15.

We introduce the two spinor representations which will be the building blocks of the Dirac space.

Definition 6.5.8 (Weyl spinors)**.** *On $\mathbb{C}^2$ we define*

$$D^{(1/2,0)}(g) := g, \qquad D^{(0,1/2)}(g) := (g^\dagger)^{-1} = \theta(g),$$

which we call the left-handed *and the* right-handed Weyl representation*. Their differentials are*

$$dD^{(1/2,0)}(X) = X, \qquad dD^{(0,1/2)}(X) = -X^\dagger.$$

By Lemma 6.1.15, $dD^{(1/2,0)}$ sends $A_k \mapsto -\frac{i}{2}\sigma_k$ and $B_k \mapsto 0$, whereas $dD^{(0,1/2)}$ sends $A_k \mapsto 0$ and $B_k \mapsto -\frac{i}{2}\sigma_k$. Thus in the first one only the factor A acts, with spin $\frac{1}{2}$, and in the second one only the factor B does. Both representations restrict to π on $SU(2)$, because $\theta|_{SU(2)} = \mathrm{id}$. By Lemma 6.5.7, $D^{(1/2,0)} \circ \theta = D^{(0,1/2)}$ and $D^{(0,1/2)} \circ \theta = D^{(1/2,0)}$.

Theorem 6.5 (Finite-dimensional representations of $SL(2,\mathbb{C})$)**.** *Every finite-dimensional continuous complex representation of $SL(2,\mathbb{C})$ is completely reducible. The irreducible ones are the representations $D^{(j_1,j_2)}$, $j_1, j_2 \in \{0, \frac{1}{2}, 1, \dots\}$, in which the A_k act as a representation of spin j_1 and the B_k as a representation of spin j_2. Moreover $D^{(j_1,j_2)}|_{SU(2)} \cong d_{j_1} \otimes d_{j_2}$ and $D^{(j_1,j_2)} \circ \theta \cong D^{(j_2,j_1)}$.*

See [**TODO-Hal03**]. The last statement follows from $d\theta(r_k) = r_k$ and $d\theta(b_k) = -b_k$, which exchange A_k and B_k.

We now show why unitarity must be abandoned. A finite-dimensional unitary representation of the Lorentz group would be the natural candidate for a local transformation law, but it does not exist.

Proposition 6.5.9 (No finite-dimensional unitary representations)**.** *Every continuous homomorphism $M : SL(2,\mathbb{C}) \to U(n)$ is trivial.*

Proof. M is smooth, and its differential maps $\mathfrak{sl}(2,\mathbb{C})$ into the skew-Hermitian matrices. Set $X_k := dM(A_k)$ and $Y_k := dM(B_k)$. Then $dM(r_k) = X_k + Y_k$ is skew-Hermitian, and $dM(b_k) = i(X_k - Y_k)$ is skew-Hermitian, *i.e.*, $X_k - Y_k$ is Hermitian. Adding the two conditions, $X_k^\dagger = -Y_k$. By Equation (6.11), $[X_j, Y_k] = 0$, that is $[X_j, X_k^\dagger] = 0$ for all j, k, and $X_3 = [X_1, X_2]$. Using $X_2 X_3^\dagger = X_3^\dagger X_2$ and cyclicity of the trace,

$$\mathrm{tr}(X_3 X_3^\dagger) = \mathrm{tr}(X_1 X_2 X_3^\dagger) - \mathrm{tr}(X_2 X_1 X_3^\dagger) = \mathrm{tr}(X_1 X_3^\dagger X_2) - \mathrm{tr}(X_2 X_1 X_3^\dagger) = 0.$$

Hence $X_3 = 0$, and by cyclic permutation $X_1 = X_2 = 0$, so $Y_k = -X_k^\dagger = 0$ and $dM = 0$. Since $SL(2,\mathbb{C})$ is connected, M is trivial. □

Proposition 6.5.10 (Obstruction to a local unitary realization)**.** *Let V' be a finite-dimensional Hilbert space and $M : SL(2,\mathbb{C}) \to GL(V')$ a continuous representation such that*

$$\big(\mathcal{U}_M(a,g)\psi\big)(p) := e^{ia\cdot p/\hbar}\, M(g)\, \psi\big(\Lambda(g)^{-1}p\big)$$

is unitary on $L^2(\mathcal{O}_m^+, \mu; V')$ for all (a, g). Then M is trivial. Consequently, for half-integer j the representation $U_{m,j}$ is not unitarily equivalent to any representation of this form.

Proof. Take $\psi = \chi_B v$, with $v \in V'$ and B a Borel set with $0 < \mu(B) < \infty$. By the invariance of μ, $\|\mathcal{U}_M(0,g)\psi\|^2 = \mu(B)\|M(g)v\|^2$, which equals $\|\psi\|^2 = \mu(B)\|v\|^2$. Hence $M(g)$ is unitary, and Proposition 6.5.9 yields $M = \mathbb{1}$. For half-integer j, $U_{m,j}(0,-\mathbb{1}) = -\mathbb{1}$ by Corollary 6.4.10, whereas $\mathcal{U}_M(0,-\mathbb{1}) = M(-\mathbb{1}) = \mathbb{1}$. □

Remark 6.5.11. Proposition 6.5.10 is the precise form of the statement that spin and momentum cannot be separated. It also explains the first move of the strategy: the matrices $D(g)$ of a local realization cannot be unitary, and therefore the Euclidean norm of W cannot be the norm of the representation. Unitarity of $\mathcal{U}_D$ will be recovered in Section 6.7 with a different, indefinite on W but positive on $F(p)$, Hermitian form.

Remark 6.5.12 (A single Weyl spinor is already local). Locality alone does not force a wave equation, nor a doubling of components. Take $W=\mathbb{C}^2$, $D=D^{(1/2,0)}$ and $\iota=\mathrm{id}$, so that $F(p)=\mathbb{C}^2$ for every p and $\Psi(p)=A_p\psi(p)$. By Lemma 6.5.4 this is a local realization of the induced representation with $j=\frac{1}{2}$, with no constraint at all. The norm of ψ is $\int \Psi^\dagger(\tilde{p}/m)\,\Psi\,d\mu$ by Equation (6.14), and it is Lorentz invariant because Equation (6.5) yields $g^\dagger\,\widetilde{\Lambda(g)p}\,g=\tilde{p}$. What a single Weyl spinor cannot do is accommodate space inversion, as we now show.

Space Inversion and the Doubling of Spinors

In the Galilean chapter no discrete symmetry was needed. Here we add a requirement which is of a different nature from Definition 6.3.1. Space inversion is not assumed to act on the state space, and it is not a consequence of the Poincaré group and of the Kähler structure, as Remark 6.3.3 records. Weak interactions show that it is not a universal law of nature. What we require is that the local realization be compatible with it, in the sense that the space of fields carries an operator implementing it by a fixed matrix. Lemma 6.5.13 below shows that such an operator exists on the induced representation of a free massive particle, for each of the two signs $\epsilon=\pm1$, and the requirement is that it can be realized locally. This is what a field theory of a parity-invariant interaction would ask for.

The consequence of Lemma 6.5.7 on the induced representation is that parity acts without Wigner rotations, and this is the property that will be transported to the field space. We verify it before turning to the fields, since it fixes the form of the parity operator that a local realization has to reproduce.

Lemma 6.5.13 (Parity on the induced representation)**.** *For $\epsilon\in\{+1,-1\}$ let $(U^\epsilon(\mathsf{P})\psi)(p):=\epsilon\,\psi(\mathsf{P}p)$ on $\mathcal{H}_{m,j}$. Then $U^\epsilon(\mathsf{P})$ is unitary, $U^\epsilon(\mathsf{P})^2=\mathbb{1}$, and*

$$U^\epsilon(\mathsf{P})\,U(a,g)\,U^\epsilon(\mathsf{P})^{-1}=U(\mathsf{P}a,\theta(g)). \tag{6.24}$$

Proof. Unitarity follows from the invariance of μ under P, item 3 of Lemma 6.5.7, and $U^\epsilon(\mathsf{P})^2=\mathbb{1}$ from $\mathsf{P}^2=\mathrm{id}$ and $\epsilon^2=1$. We first show that $R_W(\theta(g),p)=R_W(g,\mathsf{P}p)$. By Equations (6.22) and (6.23),

$$R_W(\theta(g),p)=A_p^{-1}\theta(g)A_{\mathsf{P}\Lambda(g)^{-1}\mathsf{P}p}=\theta(A_{\mathsf{P}p})^{-1}\,\theta(g)\,\theta\big(A_{\Lambda(g)^{-1}\mathsf{P}p}\big)=\theta\big(R_W(g,\mathsf{P}p)\big)=R_W(g,\mathsf{P}p),$$

where the last equality holds because $R_W(g,\mathsf{P}p)\in SU(2)$ and θ is the identity on $SU(2)$. Then, using $a\cdot\mathsf{P}p=\mathsf{P}a\cdot p$ and $\mathsf{P}\Lambda(g)^{-1}\mathsf{P}=\Lambda(\theta(g))^{-1}$,

$$\begin{aligned}\big(U^\epsilon(\mathsf{P})U(a,g)U^\epsilon(\mathsf{P})^{-1}\psi\big)(p)&=\epsilon\,e^{ia\cdot\mathsf{P}p/\hbar}\pi(R_W(g,\mathsf{P}p))\,\epsilon\,\psi\big(\mathsf{P}\Lambda(g)^{-1}\mathsf{P}p\big)\\&=e^{i\mathsf{P}a\cdot p/\hbar}\pi\big(R_W(\theta(g),p)\big)\psi\big(\Lambda(\theta(g))^{-1}p\big)=\big(U(\mathsf{P}a,\theta(g))\psi\big)(p).\end{aligned}$$ □

The two signs $\epsilon=\pm1$ are the two possible intrinsic parities. They are not distinguished by the group relations, as will be seen again in Remark 6.6.3.

Definition 6.5.14 (Parity-covariant local realization)**.** *Let (D,W,ι) be a local covariant realization. A* parity matrix *for it is a matrix $M\in GL(W)$ with $M^2=\mathbb{1}$ and*

$$M\,D(g)\,M^{-1}=D(\theta(g)),\qquad g\in SL(2,\mathbb{C}). \tag{6.25}$$

Space inversion acts on fields by $(\mathcal{U}_D(\mathsf{P})\Psi)(p):=M\,\Psi(\mathsf{P}p)$.

As in the proof of Lemma 6.5.13, Equation (6.25) and $\mathsf{P}\Lambda(g)^{-1}\mathsf{P}=\Lambda(\theta(g))^{-1}$ yield $\mathcal{U}_D(\mathsf{P})\mathcal{U}_D(a,g)\mathcal{U}_D(\mathsf{P})^{-1}=\mathcal{U}_D(\mathsf{P}a,\theta(g))$. It descends that $\mathcal{U}_D$ extends to the group generated by $\mathcal{P}$ and an involution P acting on $\mathcal{P}$ by $(a,g)\mapsto(\mathsf{P}a,\theta(g))$. There is another double covering of the orthochronous Lorentz group with inversions, in which $\mathsf{P}^2=-\mathbb{1}$, and we do not consider it.

Lemma 6.5.15 (Parity stability)**.** *Let (D, W, ι) be a local covariant realization with a parity matrix M. Then $\mathcal{F}_D$ is invariant under $\mathcal{U}_D(\mathsf{P})$ if and only if $MF(\mathring{p}) = F(\mathring{p})$. In this case $MF(p) = F(\mathsf{P}p)$ for all $p \in \mathcal{O}_m^+$.*

Proof. By Equations (6.25) and (6.23),

$$MD(A_p) = D(\theta(A_p))M = D(A_{\mathsf{P}p})M, \qquad \text{hence} \qquad MF(p) = D(A_{\mathsf{P}p})MF(\mathring{p}).$$

If $MF(\mathring{p}) = F(\mathring{p})$, then $MF(p) = F(\mathsf{P}p)$ for all p, and $M\Psi(\mathsf{P}p) \in MF(\mathsf{P}p) = F(p)$. Conversely, invariance of $\mathcal{F}_D$ means $MF(\mathsf{P}p) \subseteq F(p)$ for all p, and $p = \mathring{p}$ yields $MF(\mathring{p}) \subseteq F(\mathring{p})$, with equality because M is invertible. □

The Weyl representations cannot carry such a matrix. This is the algebraic form of the fact that parity exchanges the two chiralities.

Proposition 6.5.16 (The parity problem)**.** *1. There is no $T \in GL(2,\mathbb{C})$ with $T\,g\,T^{-1} = (g^\dagger)^{-1}$ for all $g \in SL(2,\mathbb{C})$. In particular $D^{(1/2,0)}$ and $D^{(0,1/2)}$ are not equivalent.*

2. Consequently neither $D^{(1/2,0)}$ nor $D^{(0,1/2)}$ admits a parity matrix.

Proof. Item 1. Suppose T exists. For $u \in SU(2)$, $(u^\dagger)^{-1} = u$, so T commutes with all of $SU(2)$, and $T = c\,\mathbb{1}$ by Schur's lemma (Lemma **??**), since the defining representation of $SU(2)$ on $\mathbb{C}^2$ is irreducible. Then $g = (g^\dagger)^{-1}$ for all $g \in SL(2,\mathbb{C})$, which is false for $g = \mathrm{diag}(2, \frac{1}{2})$.

Item 2. A parity matrix for $D^{(1/2,0)}$ would satisfy $MgM^{-1} = D^{(1/2,0)}(\theta(g)) = (g^\dagger)^{-1}$, which contradicts item 1. For $D^{(0,1/2)}$, Equation (6.25) reads $M(g^\dagger)^{-1}M^{-1} = g$, and $g \mapsto \theta(g)$ turns this into the same contradiction. □

The failure is a failure of equivalence between D and $D \circ \theta$, and any parity-covariant realization of a spin $\frac{1}{2}$ particle must therefore contain both chiralities. The smallest way to do so is unique.

Proposition 6.5.17 (Minimal parity-covariant space)**.** *Let (D, W, ι) be a local covariant realization of $(d_{1/2}, \mathbb{C}^2)$ with a parity matrix M. Then $\dim W \geq 4$, and $\dim W = 4$ if and only if $D \cong D^{(1/2,0)} \oplus D^{(0,1/2)}$.*

Proof. By Theorem 6.5, $D \cong \bigoplus_{(j_1,j_2)} n(j_1,j_2)\,D^{(j_1,j_2)}$ with multiplicities $n(j_1,j_2) \in \mathbb{N}$. Equation (6.25) says that M is an equivalence between D and $D \circ \theta \cong \bigoplus n(j_1,j_2)D^{(j_2,j_1)}$, so $n(j_1,j_2) = n(j_2,j_1)$. The map ι embeds $d_{1/2}$ in $D|_{SU(2)}$, so some summand (j_1,j_2) with $n(j_1,j_2) > 0$ has $d_{j_1} \otimes d_{j_2} = \bigoplus_{J=|j_1-j_2|}^{j_1+j_2} d_J \ni d_{1/2}$. This requires $|j_1 - j_2| \leq \frac{1}{2}$ and $\frac{1}{2} - |j_1 - j_2| \in \mathbb{Z}$, *i.e.*, $|j_1 - j_2| = \frac{1}{2}$. In particular $j_1 \neq j_2$, so that (j_2, j_1) is also present, and

$$\dim W \geq 2(2j_1+1)(2j_2+1) \geq 4,$$

with equality if and only if $\{j_1, j_2\} = \{0, \frac{1}{2}\}$ and W contains nothing else. □

Definition 6.5.18 (Dirac space)**.** *The* Dirac space *is $W_{\mathrm{D}} := \mathbb{C}^4 = \mathbb{C}^2 \oplus \mathbb{C}^2$ with the representation and the matrix*

$$D(g) := \begin{pmatrix} g & 0 \\ 0 & (g^\dagger)^{-1} \end{pmatrix}, \qquad \gamma^0 := \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix},$$

i.e., $D = D^{(1/2,0)} \oplus D^{(0,1/2)}$. We write $\gamma_0 = \gamma^0$ and $M := \gamma^0$.

Lemma 6.5.19. *1. $(\gamma^0)^2 = \mathbb{1}$, γ^0 is Hermitian, and $\gamma^0 D(g) \gamma^0 = D(\theta(g))$, so γ^0 is a parity matrix.*

2. $D(g)^\dagger \gamma^0 D(g) = \gamma^0$ for all $g \in SL(2,\mathbb{C})$.

3. $D(u) = \mathrm{diag}(u,u)$ is unitary for $u \in SU(2)$, and $D(A_p) = \mathrm{diag}(A_p, A_p^{-1})$.

Proof. Item 1. $\gamma^0 D(g) \gamma^0 = \mathrm{diag}((g^\dagger)^{-1}, g) = D(\theta(g))$ because $\theta(g)^\dagger = g^{-1}$. Item 2.

$$D(g)^\dagger \gamma^0 D(g) = \begin{pmatrix} g^\dagger & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} \begin{pmatrix} g & 0 \\ 0 & (g^\dagger)^{-1} \end{pmatrix} = \begin{pmatrix} 0 & g^\dagger (g^\dagger)^{-1} \\ g^{-1} g & 0 \end{pmatrix} = \gamma^0.$$

Item 3 follows from $\theta(u) = u$ and $\theta(A_p) = A_p^{-1}$, Equation (6.23). □

6.6 The Dirac Equation from the Rest Frame

Section 6.5 reduced the problem to a choice. A local covariant realization of spin $\frac{1}{2}$ which is compatible with space inversion lives on the Dirac space $\mathbb{C}^4$ by Proposition 6.5.17, and it is determined by a single object, the subspace $F(\mathring{p}) \subseteq \mathbb{C}^4$ of the physical states at rest. The two subsections below determine this subspace and transport it along the mass shell.

The first subsection uses the only symmetries left at rest, the rotations and space inversion. They leave a choice between two subspaces, which are equivalent. The second subsection transports the selected subspace with the standard boosts, and the transported constraint is the Dirac equation. No assumption on the form of the equation enters at any stage, in accordance with Remark 6.3.3. The equation is the output of covariance on the Dirac space, in the same way as the Hamiltonian of Chapter **??** was the output of covariance on the Galilean state space.

6.6.1 The Rest Subspace

From now on $j = \frac{1}{2}$, $V = \mathbb{C}^2$ and $W = W_{\mathrm{D}}$. A local covariant realization $(D, W_{\mathrm{D}}, \iota)$ is determined by the subspace $F(\mathring{p}) = \iota(V) \subseteq \mathbb{C}^4$, because all the other $F(p)$ are obtained from it by Equation (6.20) and Definition 6.5.3. In the rest frame the only symmetry left is $SU(2)$, together with space inversion, which fixes $\mathring{p}$. We determine $F(\mathring{p})$ from these two requirements, which are the only inputs.

Kinematic Decoupling

Restricted to the stabilizer, the Dirac space decomposes in a way that separates the spin from the chirality. Identify $(\xi, \zeta) \in \mathbb{C}^2 \oplus \mathbb{C}^2$ with $\xi \otimes e_1 + \zeta \otimes e_2$, where e_1, e_2 is the canonical basis of $\mathbb{C}^2_{\mathrm{mult}}$. We obtain

$$\mathbb{C}^4 \cong \mathbb{C}^2_{\mathrm{spin}} \otimes \mathbb{C}^2_{\mathrm{mult}}, \qquad D(u) = u \otimes \mathbb{1}, \qquad \gamma^0 = \mathbb{1} \otimes \sigma_1,$$

since $\gamma^0(\xi, \zeta) = (\zeta, \xi) \leftrightarrow \zeta \otimes e_1 + \xi \otimes e_2 = (\mathbb{1} \otimes \sigma_1)(\xi \otimes e_1 + \zeta \otimes e_2)$. Thus the rotations act only on the spin factor, and the parity matrix only on the multiplicity factor. This is the reason why the two requirements do not interfere.

Lemma 6.6.1 ($SU(2)$-invariant subspaces)**.** *The $SU(2)$-invariant subspaces of $\mathbb{C}^4$ are the subspaces $\mathbb{C}^2_{\mathrm{spin}} \otimes L$ with $L \subseteq \mathbb{C}^2_{\mathrm{mult}}$ a subspace, and such a subspace is isomorphic to the spin $\frac{1}{2}$ representation if and only if* $\dim L = 1$. *The invariant subspaces isomorphic to spin $\frac{1}{2}$ form a family parametrized by $L \in \mathbb{CP}^1$.*

Proof. Since $D|_{SU(2)}$ is unitary, the orthogonal projector Q onto an invariant subspace commutes with all $u \otimes \mathbb{1}$. The matrices $u \in SU(2)$ span $M_2(\mathbb{C})$, as the defining representation is irreducible (Burnside), and the commutant of $M_2(\mathbb{C}) \otimes \mathbb{1}$ is $\mathbb{1} \otimes M_2(\mathbb{C})$. Hence $Q = \mathbb{1} \otimes Q'$ with Q' a projector in $M_2(\mathbb{C})$, and the subspace is $\mathbb{C}^2 \otimes \operatorname{ran} Q'$. It is isomorphic to $(\dim L)$ copies of spin $\frac{1}{2}$. □

So the requirement that $F(\mathring{p})$ carries exactly one copy of the spin $\frac{1}{2}$ representation, as Equation (6.19) demands, leaves a whole complex projective line of possibilities. Invariance under space inversion removes all but two.

Proposition 6.6.2 (Selection of the rest subspace)**.** *Let $(D, W_{\mathrm{D}}, \iota)$ be a local covariant realization of $(d_{1/2}, \mathbb{C}^2)$ such that $\mathcal{F}_D$ is invariant under $\mathcal{U}_D(\mathsf{P})$, with $M = \gamma^0$. Then there is $\epsilon \in \{+1, -1\}$ with*

$$F(\mathring{p}) = \{\varphi \in \mathbb{C}^4 \mid \gamma^0 \varphi = \epsilon \varphi\} = \{(\eta, \epsilon\eta) \mid \eta \in \mathbb{C}^2\}, \tag{6.26}$$

ι is a multiple of $\iota_\epsilon \eta := (\eta, \epsilon\eta)$, and, with $\iota = \iota_\epsilon$,

$$J\, U^\epsilon(\mathsf{P}) = \mathcal{U}_D(\mathsf{P}) J.$$

Conversely, for each ϵ the triple $(D, W_{\mathrm{D}}, \iota_\epsilon)$ is a local covariant realization for which $\mathcal{F}_D$ is parity invariant.

Proof. By Equation (6.19) and Lemma 6.6.1, $F(\mathring{p}) = \mathbb{C}^2 \otimes L$ with $\dim L = 1$. By Lemma 6.5.15, $\gamma^0 F(\mathring{p}) = F(\mathring{p})$, that is $(\mathbb{1} \otimes \sigma_1)(\mathbb{C}^2 \otimes L) = \mathbb{C}^2 \otimes \sigma_1 L = \mathbb{C}^2 \otimes L$, so $\sigma_1 L = L$. The matrix σ_1 has simple spectrum $\{\pm 1\}$, so L is one of its two eigenlines $\mathbb{C}(e_1 + \epsilon e_2)$. Hence $F(\mathring{p}) = \{\eta \otimes (e_1 + \epsilon e_2)\} = \{(\eta, \epsilon\eta)\} = \ker(\gamma^0 - \epsilon)$. By Schur's lemma the intertwiner $\iota : V \to F(\mathring{p})$ is unique up to a scalar, so it is a multiple of ι_ϵ. For the last identity, Equation (6.23) and Equation (6.25) yield $D(A_{\mathsf{P}p}) = \gamma^0 D(A_p)\gamma^0$, and $\gamma^0 \iota_\epsilon = \epsilon \iota_\epsilon$, so

$$(\mathcal{U}_D(\mathsf{P})J\psi)(p) = \gamma^0 D(A_{\mathsf{P}p})\iota_\epsilon \psi(\mathsf{P}p) = D(A_p)\gamma^0 \iota_\epsilon \psi(\mathsf{P}p) = D(A_p)\iota_\epsilon\, \epsilon\psi(\mathsf{P}p) = (J\, U^\epsilon(\mathsf{P})\psi)(p).$$

The converse is a direct check: ι_ϵ satisfies Equation (6.19) since $D(u) = u \otimes \mathbb{1}$, and $\gamma^0 F(\mathring{p}) = F(\mathring{p})$ by construction. □

Remark 6.6.3 (Intrinsic parity and the choice of the sign). The sign ϵ is the intrinsic parity of Lemma 6.5.13: space inversion acts on the amplitude ψ as $\epsilon\psi(\mathsf{P}p)$. The two choices are equivalent. The matrix $\gamma_5 := \mathrm{diag}(\mathbb{1}, -\mathbb{1})$ commutes with every $D(g)$, anticommutes with γ^0, and maps $\ker(\gamma^0 - \epsilon)$ onto $\ker(\gamma^0 + \epsilon)$. Hence $\Psi \mapsto \gamma_5 \Psi$ intertwines the two realizations $\mathcal{U}_D$ and exchanges the parity matrices M and $-M$, and $-M$ is as good as M. Since only relative intrinsic parities are observable, we fix $\epsilon = +1$ from now on.

Definition 6.6.4 (Rest projector)**.** *With $\epsilon = +1$, $F(\mathring{p}) = \{(\eta, \eta) \mid \eta \in \mathbb{C}^2\}$, $\iota\eta := (\eta, \eta)$, and the* rest projector *is the orthogonal projector of $\mathbb{C}^4$ onto $F(\mathring{p})$,*

$$\mathcal{Q}(\mathring{p}) := \tfrac{1}{2}(\mathbb{1} + \gamma^0).$$

Summing up, the group theory fixes everything except one complex line L and, once the Dirac space is chosen, invariance under space inversion fixes L up to the sign ϵ. No further input on the form of the equation is used.

6.6.2 Covariant Transport and the Dirac Equation

The rest subspace $F(\mathring{p}) = \{(\eta, \eta)\}$ is now fixed, and the constraint at an arbitrary momentum is obtained by transport with the standard boost, $F(p) = D(A_p)F(\mathring{p})$. Two things must be checked: that the transport does not depend on the choice of the boost, and that the resulting family of subspaces is stable under the whole symmetry group. Only then can the transported constraint be written as an equation.

Proposition 6.6.5 (Coherence of the constraint)**.** *Let $F(\mathring{p}) = \{(\eta, \eta) \mid \eta \in \mathbb{C}^2\}$ and $F(p) = D(A_p)F(\mathring{p})$ for $p \in \mathcal{O}_m^+$.*

1. *If $A' \in SL(2, \mathbb{C})$ satisfies $\Lambda(A')\mathring{p} = p$, then $D(A')F(\mathring{p}) = F(p)$.*
2. *$D(g)F(p) = F(\Lambda(g)p)$ for all $g \in SL(2, \mathbb{C})$.*
3. *$\gamma^0 F(p) = F(\mathsf{P}p)$.*
4. *$\dim F(p) = 2$.*

Proof. Item 1. By Proposition 6.4.1, $A' = A_p u$ with $u \in SU(2)$, and $D(u)(\eta, \eta) = (u\eta, u\eta)$, so $D(A')F(\mathring{p}) = D(A_p)D(u)F(\mathring{p}) = D(A_p)F(\mathring{p})$. Thus $F(p)$ does not depend on the choice of the boost, because the rest subspace is $SU(2)$-invariant.

Item 2. Let $q := \Lambda(g)p$. Then $gA_p = A_q u$ with $u = A_q^{-1} g A_p = R_W(g, q) \in SU(2)$, and $D(g)F(p) = D(A_q)D(u)F(\mathring{p}) = F(q)$. This is the stability of the constraint under the Wigner rotations: the rotation u moves $F(\mathring{p})$ into itself.

Item 3 is Lemma 6.5.15, applicable since $\gamma^0 F(\mathring{p}) = F(\mathring{p})$. Item 4 holds because $D(A_p)$ is invertible. □

Item 2 says that the family $F(p)$ is an $SL(2, \mathbb{C})$-invariant subbundle of the trivial bundle $\mathcal{O}_m^+ \times \mathbb{C}^4$, in the sense of Remark 6.5.5, and item 1 that it is determined by its fibre at the rest momentum.

Item 3 uses $A_{\mathsf{P}p} = \theta(A_p)$, *i.e.*, that space inversion exchanges A_p with A_p^{-1}, see Equation (6.23). We now compute the transported projector. The gamma matrices appear here as the images of the Hermitian matrix model.

Lemma 6.6.6 (Gamma matrices)**.** *Let γ^0 be as in Definition 6.5.18 and $\gamma^k := \begin{pmatrix} 0 & -\sigma_k \\ \sigma_k & 0 \end{pmatrix}$ for $k = 1,2,3$. For $x \in \mathbb{M}$ set $\gamma(x) := \gamma_\mu x^\mu = \gamma^0 x^0 - \sum_k \gamma^k x^k$. Then*

1. *$\gamma(x) = \begin{pmatrix} 0 & \hat{x} \\ \tilde{x} & 0 \end{pmatrix}$, $\gamma(x)^2 = x^2\,\mathbb{1}$, hence $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{1}$ with $\eta = \mathrm{diag}(1,-1,-1,-1)$, and $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$.*
2. *$D(g)\,\gamma(x)\,D(g)^{-1} = \gamma(\Lambda(g)x)$ for all $g \in SL(2,\mathbb{C})$.*
3. *$\gamma^0\gamma(x)\gamma^0 = \gamma(\mathsf{P}x)$.*

Proof. Item 1. The block form is read off from the definitions, and $\gamma(x)^2 = \mathrm{diag}(\hat{x}\tilde{x}, \tilde{x}\hat{x}) = x^2\mathbb{1}$ by Lemma 6.1.2. The Clifford relations follow by polarization. For the last claim, $\gamma^{0\dagger} = \gamma^0$, while $\gamma^{k\dagger} = \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix} = \gamma^0\gamma^k\gamma^0$.

Item 2. By Equation (6.5),

$$D(g)\gamma(x)D(g)^{-1} = \begin{pmatrix} g & 0 \\ 0 & (g^\dagger)^{-1} \end{pmatrix}\begin{pmatrix} 0 & \hat{x} \\ \tilde{x} & 0 \end{pmatrix}\begin{pmatrix} g^{-1} & 0 \\ 0 & g^\dagger \end{pmatrix} = \begin{pmatrix} 0 & g\hat{x}g^\dagger \\ (g^\dagger)^{-1}\tilde{x}g^{-1} & 0 \end{pmatrix} = \gamma(\Lambda(g)x).$$

Item 3. $\gamma^0\gamma(x)\gamma^0 = \begin{pmatrix} 0 & \tilde{x} \\ \hat{x} & 0 \end{pmatrix}$, and $\widehat{\mathsf{P}x} = \tilde{x}$, $\widetilde{\mathsf{P}x} = \hat{x}$. □

Proposition 6.6.7 (Transport of the rest projector)**.** *For every $p \in \mathcal{O}_m^+$,*

$$D(A_p)\,\gamma^0\,D(A_p)^{-1} = \frac{\gamma_\mu p^\mu}{m}. \tag{6.27}$$

Proof. Explicitly, using Lemma 6.5.19 and Equation (6.14),

$$\begin{pmatrix} A_p & 0 \\ 0 & A_p^{-1} \end{pmatrix}\begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}\begin{pmatrix} A_p^{-1} & 0 \\ 0 & A_p \end{pmatrix} = \begin{pmatrix} 0 & A_p^2 \\ A_p^{-2} & 0 \end{pmatrix} = \frac{1}{m}\begin{pmatrix} 0 & \hat{p} \\ \tilde{p} & 0 \end{pmatrix} = \frac{\gamma(p)}{m}.$$

The same follows from item 2 of Lemma 6.6.6, since $\gamma^0 = \gamma(\mathring{p})/m$ and $\Lambda(A_p)\mathring{p} = p$. □

Equation (6.27) is the tensor transport announced in the introduction: the rest projector, which singles out the chirality-symmetric combination at rest, becomes the four-momentum contracted with the gamma matrices. The equation of the chapter follows.

Theorem 6.6 (Dirac equation in momentum space)**.** *For $p \in \mathcal{O}_m^+$ let $\mathcal{Q}(p) := D(A_p)\,\mathcal{Q}(\mathring{p})\,D(A_p)^{-1}$. Then*

$$\mathcal{Q}(p) = \tfrac{1}{2}\Big(\mathbb{1} + \frac{\gamma_\mu p^\mu}{m}\Big) \tag{6.28}$$

is a projector of rank 2 with range $F(p)$, and

$$F(p) = \big\{\Psi \in \mathbb{C}^4 \;\big|\; (\gamma_\mu p^\mu - m)\Psi = 0\big\}. \tag{6.29}$$

Consequently the space of fields of Definition 6.5.3 is

$$\mathcal{F}_D = \big\{\Psi : \mathcal{O}_m^+ \to \mathbb{C}^4 \;\big|\; (\gamma_\mu p^\mu - m)\Psi(p) = 0 \ \text{ for all } p\big\},$$

and, by Lemma 6.5.4, $\mathcal{U}_D$ realizes the induced representation $U_{m,1/2}$, with the intrinsic parity $\epsilon = +1$.

Proof. Equation (6.28) is Proposition 6.6.7. The matrix $\mathcal{Q}(p)$ is conjugate to the projector $\mathcal{Q}(\mathring{p})$, hence it is a projector of rank 2, and its range is $D(A_p)\,\mathrm{ran}\,\mathcal{Q}(\mathring{p}) = D(A_p)F(\mathring{p}) = F(p)$. Moreover $\mathcal{Q}(p)\Psi = \Psi$ if and only if $\gamma(p)\Psi = m\Psi$. □

Remark 6.6.8. The projector $\mathcal{Q}(p)$ is not Hermitian for $\mathbf{p} \neq \mathbf{0}$, since $\gamma(p)^\dagger = \begin{pmatrix} 0 & \tilde{p} \\ \hat{p} & 0 \end{pmatrix} \neq \gamma(p)$. This reflects the fact that $D(A_p)$ is not unitary, and it is the same phenomenon which forced us to abandon unitarity in Proposition 6.5.10. The consequences for the norm are drawn in Section 6.7.

The equation in position space is obtained from Remark 6.5.6. Since the plane wave $e^{-ip\cdot x/\hbar}$ satisfies $i\hbar\partial_\mu e^{-ip\cdot x/\hbar} = p_\mu e^{-ip\cdot x/\hbar}$, multiplication by $\gamma_\mu p^\mu$ becomes a first-order differential operator.

Corollary 6.6.9 (Dirac equation in position space)**.** *Let $\Psi \in \mathcal{F}_D$ be such that $\Psi(x) = \int e^{-ip\cdot x/\hbar}\Psi(p)\,d\mu(p)$ is well defined together with its derivatives, for instance continuous with compact support. Then*

$$\big(i\hbar\,\gamma^\mu\partial_\mu - m\big)\Psi(x) = 0, \tag{6.30}$$

i.e., $i\hbar\,\partial_t\Psi = \big(\boldsymbol{\alpha}\cdot(-i\hbar\nabla) + m\beta\big)\Psi$ with $\beta := \gamma^0$ and $\alpha^k := \gamma^0\gamma^k = \mathrm{diag}(\sigma_k, -\sigma_k)$.

Proof. Apply $i\hbar\gamma^\mu\partial_\mu$ under the integral to get $\int \gamma_\mu p^\mu e^{-ip\cdot x/\hbar}\Psi(p)\,d\mu(p) = m\Psi(x)$, by Equation (6.29). Multiplying by γ^0 and using $(\gamma^0)^2 = \mathbb{1}$ yields the Hamiltonian form, with $\mathbf{p} \to -i\hbar\nabla$ and $p^0 \to i\hbar\partial_t$. □

The Hamiltonian form closes the loop with Theorem 6.3. The matrix $h_{\mathrm{D}}(\mathbf{p}) := \boldsymbol{\alpha}\cdot\mathbf{p} + m\beta$ is the Dirac Hamiltonian of a plane wave, and the constraint selects precisely its positive spectral subspace.

Proposition 6.6.10 (The constraint is the positive spectral subspace)**.** *$h_{\mathrm{D}}(\mathbf{p})$ is Hermitian with $h_{\mathrm{D}}(\mathbf{p})^2 = \omega(\mathbf{p})^2\,\mathbb{1}$, and $F(p) = \ker\big(h_{\mathrm{D}}(\mathbf{p}) - \omega(\mathbf{p})\big)$. In particular the energy $p^0 = \omega(\mathbf{p})$ of Theorem 6.3 is the eigenvalue of the Dirac Hamiltonian on $F(p)$.*

Proof. The matrices α^k and β are Hermitian and, by the Clifford relations of Lemma 6.6.6, $\{\alpha^j, \alpha^k\} = 2\delta_{jk}\mathbb{1}$, $\{\alpha^k, \beta\} = 0$, $\beta^2 = \mathbb{1}$, so that $h_{\mathrm{D}}^2 = (|\mathbf{p}|^2 + m^2)\mathbb{1}$. Since $\mathrm{tr}\,\alpha^k = \mathrm{tr}\,\beta = 0$, h_{D} has the eigenvalues $\pm\omega$, each with multiplicity 2. A solution of Equation (6.29) satisfies $p^0\Psi = h_{\mathrm{D}}\Psi$ with $p^0 = \omega(\mathbf{p}) > 0$, so $F(p) \subseteq \ker(h_{\mathrm{D}} - \omega)$, and the dimensions coincide. □

6.7 Probability Conservation and the Metric Structure

Section 6.5 showed that the matrices $D(g)$ of a local realization cannot be unitary. The Euclidean norm of $\mathbb{C}^4$ is therefore not the norm of the induced representation, and the question is which positive form on $F(p)$ is invariant. This section computes the solutions explicitly, exhibits the discrepancy between the Euclidean and the covariant norm, and proves that the covariant norm reproduces the scalar product of $\mathcal{H}_{m,1/2}$.

Proposition 6.7.1 (Explicit solutions)**.** *For $p \in \mathcal{O}_m^+$,*

$$F(p) = \big\{\Psi = (A_p\eta,\ A_p^{-1}\eta)^T \;\big|\; \eta \in \mathbb{C}^2\big\}.$$

For an amplitude ψ as in Lemma 6.5.4, $(J\psi)(p) = (A_p\psi(p), A_p^{-1}\psi(p))^T$, i.e., $\eta = \psi(p)$.

Proof. $F(p) = D(A_p)\{(\eta,\eta)\} = \{(A_p\eta, A_p^{-1}\eta)\}$ by Lemma 6.5.19 and Definition 6.6.4. The formula for J is Lemma 6.5.4 with $\iota\eta = (\eta,\eta)$. One checks directly that these vectors solve Equation (6.29): $\gamma(p)\Psi = (\hat{p}A_p^{-1}\eta, \tilde{p}A_p\eta) = (mA_p\eta, mA_p^{-1}\eta) = m\Psi$. □

Two quadratic forms on $\mathbb{C}^4$ are available. The Euclidean one is $\Psi^\dagger\Phi$. The other is the Dirac form $\bar{\Psi}\Phi := \Psi^\dagger\gamma^0\Phi$, where $\bar{\Psi} := \Psi^\dagger\gamma^0$ is the Dirac adjoint, not the complex conjugate. By item 2 of Lemma 6.5.19, the second one is invariant under the representation,

$$\overline{D(g)\Psi}\,D(g)\Phi = \Psi^\dagger D(g)^\dagger\gamma^0 D(g)\Phi = \bar{\Psi}\Phi, \qquad g \in SL(2,\mathbb{C}), \tag{6.31}$$

whereas $D(g)^\dagger D(g) \neq \mathbb{1}$ in general, so that the first is not.

Proposition 6.7.2 (Euclidean and covariant norm)**.** *Let $\Psi = (A_p\eta, A_p^{-1}\eta)$ and $\Phi = (A_p\eta', A_p^{-1}\eta')$ be in $F(p)$.*

1. *$\bar{\Psi}\Phi = 2\,(\eta,\eta')$. In particular the Dirac form is positive definite on $F(p)$, and $\bar{\Psi}\Psi = 2|\eta|^2$.*

2. *For every* $q \in \mathbb{M}$, $\bar\Psi\gamma(q)\Psi = \dfrac{2\,p\cdot q}{m}\,|\eta|^2$, *i.e.,* $\bar\Psi\gamma^\mu\Psi = \dfrac{2p^\mu}{m}|\eta|^2$. *In particular*

$$\Psi^\dagger\Psi = \frac{2p^0}{m}\,|\eta|^2 = \frac{p^0}{m}\,\bar\Psi\Psi. \tag{6.32}$$

Proof. Item 1. Since $\gamma^0\Phi = (A_p^{-1}\eta', A_p\eta')$, $\bar\Psi\Phi = \eta^\dagger A_p A_p^{-1}\eta' + \eta^\dagger A_p^{-1}A_p\eta' = 2(\eta,\eta')$.

Item 2. Since $\gamma^0\gamma(q) = \operatorname{diag}(\tilde q, \hat q)$, $\bar\Psi\gamma(q)\Psi = \eta^\dagger\big(A_p\tilde q A_p + A_p^{-1}\hat q A_p^{-1}\big)\eta$. Since A_p is Hermitian, Equation (6.5) and the definition of Λ with $g = A_p^{-1}$ yield $A_p\tilde q A_p = \widetilde{q'}$ and $A_p^{-1}\hat q A_p^{-1} = \widehat{q'}$ with $q' := \Lambda(A_p)^{-1}q$. Hence the bracket equals $\widetilde{q'} + \widehat{q'} = 2q'^0\,\mathbb{1}$, and $q'^0 = q\cdot\Lambda(A_p)e_0 = q\cdot p/m$. Taking $q = e_0$, for which $\gamma(e_0) = \gamma^0$, yields Equation (6.32). □

Equation (6.32) is the discrepancy between the two metrics: the Euclidean density exceeds the invariant one by the factor p^0/m, which is the time dilation factor of a particle of momentum p. It is not a defect of the construction. The Euclidean density $\Psi^\dagger\Psi$ is the time component of the four-vector $\bar\Psi\gamma^\mu\Psi$, whereas the Dirac density $\bar\Psi\Psi$ is a scalar. The covariant norm is the one built from the scalar.

Theorem 6.7 (Covariant norm and equivalence with the induced representation)**.** *On* $\mathcal{F}_D$ *let*

$$\langle\Psi,\Phi\rangle_D := \frac{1}{2}\int \bar\Psi(p)\,\Phi(p)\,d\mu(p), \qquad \mathcal{F}_D^2 := \Big\{\Psi\in\mathcal{F}_D \;\Big|\; \int \bar\Psi\Psi\,d\mu < \infty\Big\}.$$

Then $\langle\cdot,\cdot\rangle_D$ *is a positive definite scalar product,* $\mathcal{F}_D^2$ *is a Hilbert space, the map* $J : \mathcal{H}_{m,1/2} \to \mathcal{F}_D^2$ *is unitary, and* $\mathcal{U}_D(a,g)$ *is unitary on* $\mathcal{F}_D^2$ *with* $J\,U_{m,1/2}(a,g) = \mathcal{U}_D(a,g)J$. *Hence* $\mathcal{U}_D$ *is irreducible and unitarily equivalent to* $U_{m,1/2}$.

Proof. By Proposition 6.7.2, $\frac{1}{2}\overline{J\psi}(p)(J\phi)(p) = (\psi(p),\phi(p))_{\mathbb{C}^2}$ pointwise, so $\langle J\psi, J\phi\rangle_D = \langle\psi,\phi\rangle_{\mathcal{H}_{m,1/2}}$. Since J is a bijection onto $\mathcal{F}_D$ (Lemma 6.5.4), $\langle\cdot,\cdot\rangle_D$ is a scalar product making $\mathcal{F}_D^2$ a Hilbert space and J unitary. The intertwining relation is Equation (6.21). Irreducibility follows from Theorem 6.4. Directly, unitarity of $\mathcal{U}_D$ also follows from Equation (6.31) and the invariance of μ: $\overline{(\mathcal{U}_D(a,g)\Psi)(p)}(\mathcal{U}_D(a,g)\Phi)(p) = \bar\Psi(\Lambda(g)^{-1}p)\Phi(\Lambda(g)^{-1}p)$. □

Theorem 6.7 says that the scalar product of the induced representation, which was the starting point of Section 6.4, is preserved by the construction of the Dirac field: the unitary structure is not lost by the passage to a non-unitary D, it is carried by the form $\frac{1}{2}\bar\Psi\Phi$ instead of the Euclidean one.

Remark 6.7.3 (Dirac solutions exhaust the elementary systems of spin one half). Let Φ be an elementary relativistic system of mass $m > 0$ whose spin, in the sense of Theorem 6.4, is $j = \frac{1}{2}$. The unitary operator $\mathcal{W} : \mathcal{H} \to \mathcal{H}_{m,1/2}$ of that theorem and the unitary operator $J : \mathcal{H}_{m,1/2} \to \mathcal{F}_D^2$ of Theorem 6.7 compose to a unitary operator $J\mathcal{W} : \mathcal{H} \to \mathcal{F}_D^2$ with $J\mathcal{W}\Pi(\kappa) = \mathcal{U}_D(\kappa)J\mathcal{W}$ for all $\kappa \in \mathcal{P}$. It descends that the Dirac equation, with the scalar product $\langle\cdot,\cdot\rangle_D$, describes every elementary system of mass m and spin $\frac{1}{2}$, and only these, because the spin was derived from the action and not assumed. This is the precise sense in which the equation is recovered in the chapter. The equivalence concerns the representation of the symmetry group, hence the dynamics on $\mathbb{P}\mathcal{H}$, and it does not introduce a position observable, as Remark 6.7.5 explains.

The Euclidean density has a physical meaning of its own, which is the time component of a conserved current. We verify that it is conserved and that its integral is a constant multiple of the covariant norm.

Proposition 6.7.4 (Conservation of probability)**.** *Let* $\Psi \in \mathcal{F}_D$ *be smooth with compact support in* $\mathbf{p}$*, and let* $\Psi(x) = \int e^{-ip\cdot x/\hbar}\Psi(p)\,d\mu(p)$ *be the position-space field. Then* $j^\mu := \bar\Psi\gamma^\mu\Psi$ *satisfies*

$$\partial_\mu j^\mu = 0, \qquad j^0 = \Psi^\dagger\Psi \geq 0, \tag{6.33}$$

and for every $t \in \mathbb{R}$, *with* $\psi = J^{-1}\Psi$,

$$\int_{\mathbb{R}^3} \Psi^\dagger(t,\mathbf{x})\Psi(t,\mathbf{x})\,d^3\mathbf{x} = \frac{(2\pi\hbar)^3}{m}\,\|\psi\|^2_{\mathcal{H}_{m,1/2}}. \tag{6.34}$$

Proof. Let $\Psi = \Psi(x)$. By Corollary 6.6.9, $i\hbar\gamma^\mu\partial_\mu\Psi = m\Psi$. Taking the adjoint and using $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$, then multiplying on the right by γ^0, we get $-i\hbar\,\partial_\mu\bar\Psi\gamma^\mu = m\bar\Psi$. Therefore

$$\partial_\mu(\bar\Psi\gamma^\mu\Psi) = \big(\partial_\mu\bar\Psi\gamma^\mu\big)\Psi + \bar\Psi\big(\gamma^\mu\partial_\mu\Psi\big) = \frac{im}{\hbar}\bar\Psi\Psi - \frac{im}{\hbar}\bar\Psi\Psi = 0.$$

For Equation (6.34), at time t we have $\Psi(t,\mathbf{x}) = \int d^3\mathbf{p}\, e^{i(\mathbf{p}\cdot\mathbf{x}-\omega t)/\hbar}\,\Psi(p)/2\omega$, which is a Fourier transform multiplied by a phase of modulus one. By Plancherel and Equation (6.32),

$$\int \Psi^\dagger\Psi\, d^3\mathbf{x} = (2\pi\hbar)^3 \int \frac{\Psi^\dagger\Psi}{4\omega^2}\, d^3\mathbf{p} = (2\pi\hbar)^3 \int \frac{1}{4\omega^2}\frac{2\omega}{m}|\psi|^2\, d^3\mathbf{p} = \frac{(2\pi\hbar)^3}{m}\int |\psi|^2 \frac{d^3\mathbf{p}}{2\omega},$$

which is the right-hand side of Equation (6.34). □

Remark 6.7.5. The time component $j^0 = \Psi^\dagger\Psi$ of the conserved current is positive, and its integral is a multiple of the Wigner norm, so the Euclidean L^2 norm of the Dirac field in position space is conserved and is preserved by the Poincaré action, as it must be. The factor $2\omega/m$ of Equation (6.32) is exactly compensated by the Jacobian $1/4\omega^2$ between the invariant measure and the Fourier measure. This does not make $j^0(x)$ the probability density of a position measurement of the particle: the construction introduces no position observable. In the Galilean case $\mathbf{K}/m$ played this role (Remark **??**), and the corresponding relativistic question, the localization of a relativistic particle, is not touched here. Moreover the positive energy hypothesis restricts Ψ to $\mathcal{O}_m^+$. Solutions supported on $\mathcal{O}_m^-$ would yield a second irreducible representation, the one of the antiparticle, and they do not appear in $\mathcal{F}_D$.

6.8 Summary of the Construction

The chain of the chapter can be read as a sequence of questions, each answered by a symmetry requirement.

Remark 6.8.1 (What was used). *The setting.* The pure states are the points of $\mathbb{P}\mathcal{H}$, and the symmetries form an irreducible Poincaré action of positive energy (Definitions 6.3.1, 6.3.2 and 6.3.8). The projective space does not carry the spin (Remark 6.3.4), which is therefore an output.

The energy. The vanishing of $H^2(\mathfrak{p},\mathbb{R})$ (Theorem 6.2) yields a unique lift of the action, and irreducibility and positivity of the energy then yield $H = \sqrt{|\mathbf{P}|^2 + m^2}$, with the mass as a Casimir eigenvalue and no additive constant (Theorem 6.3), and the nonrelativistic limit of Proposition 6.3.16.

The representation. The vanishing of the second cohomology and the covariance of the spectral measure place the spectrum on the mass shell, whose stabilizer is $SU(2)$. Irreducibility forces the action of the stabilizer on the multiplicity space to be an irreducible representation, whose label j is the spin. The resulting representation is induced, its boosts act on the spin through the Wigner rotation (Proposition 6.4.7), and its central sign is $(-1)^{2j}$ (Corollary 6.4.10 and Theorem 6.4).

The locality. A fixed matrix transformation law requires a finite-dimensional, necessarily non-unitary, representation of $SL(2,\mathbb{C})$ (Propositions 6.5.9 and 6.5.10), and invariance under space inversion forces $D^{(1/2,0)} \oplus D^{(0,1/2)}$ (Propositions 6.5.16 and 6.5.17).

The equation. Invariance of the rest subspace under $SU(2)$ and under space inversion fixes it up to the sign of the intrinsic parity (Proposition 6.6.2), and its transport with the standard boosts is the Dirac equation (Theorem 6.6).

The norm. The scalar product of the induced representation is carried by the invariant form $\frac{1}{2}\bar\Psi\Phi$, and probability is conserved (Theorem 6.7 and Proposition 6.7.4).

No postulate on the form of the wave equation, on the number of components or on the gamma matrices has been used. The only requirement beyond the setting of Section 6.3 is the compatibility of the local realization with space inversion, which enters in Section 6.5.

Remark 6.8.2 (Outlook). The strategy of Chapter **??** and of this chapter leaves two natural continuations, which we only sketch. The first is the geometric reading, on $\mathbb{P}\mathcal{H}^\infty$, of the mean value functions of the Poincaré generators, with the Poisson algebra of Proposition **??** replaced by one in which the mass is a Casimir function and not a constant. The second is the massless case, in which the stabilizer of the momentum is not $SU(2)$ and the Weyl equation and the helicity appear in place of the Dirac equation and the spin. Interactions, and in particular minimal coupling to an electromagnetic potential, require the local transformation law of Section 6.5 and are not treated here.

Chapter 7

Induced Representations and the Geometry of Fibre Bundles

The construction of Section 6.4 introduces the space $\mathcal{E}_\pi$ of equivariant functions on $SL(2,\mathbb{C})$ and the representation $\mathcal{L}$ by left translation, and it obtains the transformation law (6.17) through the Wigner rotation $R_W(g,p)$. To a reader who has not met induced representations before, the definition of $\mathcal{E}_\pi$ may look like a formal device without geometric content. The purpose of this appendix is to show that it is the canonical formulation of a standard object of differential geometry, namely the space of sections of a vector bundle associated to a principal bundle, and that the Wigner rotation is the transition function of a trivialization of that bundle. The exposition follows the logic of the chapter and uses its notation throughout.

7.1 The General Problem

Let G be a Lie group, let $H \leq G$ be a closed subgroup and let (π, V) be a finite-dimensional unitary representation of H. We ask for a canonical way of constructing a representation of G from (π, V). The question is motivated by the chapter, where $G = SL(2,\mathbb{C})$, $H = SU(2)$ and (π, V) is the spin-j representation of the stabilizer of the rest momentum $\mathring{p}$.

The simplest attempt rests on a decomposition of the group. Suppose that $G = K \rtimes H$ for a normal subgroup $K \trianglelefteq G$, so that every $g \in G$ is uniquely $g = ku$ with $k \in K$ and $u \in H$. If $g' = k'u'$, then $gg' = k(uk'u^{-1})uu'$ with $uk'u^{-1} \in K$, hence the projection $g \mapsto u$ is a homomorphism $G \to H$, and $g \mapsto \pi(u)$ is a representation of G on V. This is what happens for the Galilei group, where the boosts and the translations form a normal subgroup and the rotations act on the spin alone, as in Corollary **??**.

The attempt fails for $G = SL(2,\mathbb{C})$ and $H = SU(2)$. The polar decomposition of Lemma 6.1.3 writes every g uniquely as $g = \exp\left(\frac{1}{2}\mathbf{w}\cdot\boldsymbol{\sigma}\right)u$, but the boosts do not form a subgroup, since the product of two boosts with different directions contains a rotation by Example 6.1.12. The decomposition is a decomposition of manifolds and not of groups, and no homomorphism of G onto H is available.

The construction of induced representations bypasses the problem. It abandons the search for a decomposition of G and works on the whole group, at the price that the representation acts on a space of functions and not on V itself. The gain is that the construction is canonical, since it requires no choice of a complement to H.

Remark 7.1.1. The representation to be built for a relativistic particle of mass m and spin j must be compatible with the full Poincaré group, which contains the translations as well. The induced representation provides the Lorentz part, and the translations are added through the semidirect structure of Definition 6.1.7, as in Definition 6.4.8.

7.2 The Principal Bundle and its Associated Vector Bundle

The geometric background of the construction is the theory of principal bundles, which we recall in the form needed here. We first fix the terminology for the pair (G,H), and we then show that the quotient G/H is the base of such a bundle.

Definition 7.2.1 (Principal bundle of a pair)**.** *Let G be a Lie group and let $H \leq G$ be a closed subgroup. The* projection

$$\varpi : G \longrightarrow G/H, \qquad \varpi(g) := gH, \tag{7.1}$$

takes values in the set of left cosets. The group G is the total space*, G/H is the* base *and the right multiplication $G \times H \to G$, $(g,h) \mapsto gh$, is the* structure action*. Its orbits are the* fibres *$\varpi^{-1}(gH) = gH$.*

Proposition 7.2.2 ($G \to G/H$ is a principal bundle)**.** *In the setting of Definition 7.2.1, G/H has a unique smooth manifold structure for which ϖ is a submersion. The structure action is free and transitive on each fibre, it commutes with the left action of G on itself, and ϖ admits smooth local sections, so that ϖ is a principal H-bundle.*

See [**TODO-Lee13**]. The two properties of the structure action, freeness and commutation with the left action, are what make the quotient construction below well defined.

In the setting of the chapter the bundle is globally trivial. The base is the mass shell of Definition 6.3.10, the identification is item 3 of Proposition 6.4.1, and the standard boosts provide a global smooth section,

$$SU(2) \;\hookrightarrow\; SL(2,\mathbb{C}) \xrightarrow{\;\varpi\;} \mathcal{O}_m^+ \cong SL(2,\mathbb{C})/SU(2). \tag{7.2}$$

We now attach a vector space to each point of the base, using the representation (π, V) of the structure group.

Definition 7.2.3 (Associated vector bundle)**.** *Let (π, V) be a representation of H. The* associated vector bundle *is the quotient*

$$E := G \times_H V := (G \times V)/\!\sim, \tag{7.3}$$

where $\sim$ is the equivalence relation generated by

$$(gh, v) \sim (g, \pi(h)v), \qquad g \in G,\ h \in H,\ v \in V. \tag{7.4}$$

The class of (g,v) is written $[g,v]$, and the projection is $\varpi_E : E \to G/H$, $\varpi_E([g,v]) := gH$.

The relation is compatible with the projection, since $(gh)H = gH$. The next lemma records that the quotient has the structure announced in its name.

Lemma 7.2.4 (The associated bundle is a vector bundle)**.** *The projection $\varpi_E : E \to G/H$ makes E a smooth vector bundle with fibre V and structure group H. For every $g \in G$ the map $v \mapsto [g,v]$ is a linear isomorphism of V onto the fibre $E_{gH} := \varpi_E^{-1}(gH)$.*

Proof. If $[g,v] = [g,v']$, the element $h \in H$ relating the two representatives satisfies $gh = g$, hence $h = e$ and $v = v'$. Therefore $v \mapsto [g,v]$ is injective, and it is surjective onto E_{gH} because every class over gH has a representative of the form (g,v) by Equation (7.4). It is linear since the relation is linear in v, and this equips each fibre with a vector space structure which does not depend on the representative g. Let $\sigma : U \to G$ be a smooth local section of ϖ, which exists by Proposition 7.2.2. Then $(p,v) \mapsto [\sigma(p), v]$ is a bijection of $U \times V$ onto $\varpi_E^{-1}(U)$, and it defines the local trivialization. If $\sigma' = \sigma h$ is another section, with $h : U \cap U' \to H$ smooth, then $[\sigma'(p), v] = [\sigma(p), \pi(h(p))v]$, so that the transition functions are $p \mapsto \pi(h(p))$, which take values in $\pi(H)$. □

In the case of Equation (7.2), one global section is enough, and $E = SL(2,\mathbb{C}) \times_{SU(2)} \mathbb{C}^{2j+1}$ is trivial. Nevertheless the group $SL(2,\mathbb{C})$ acts on it in a way that is not the trivial one, and this is the content of the next sections.

7.3 Sections and the Equivariance Condition

The space $\mathcal{E}_\pi$ of Definition 6.4.5 is not an arbitrary construction. It is the space of sections of the associated bundle, described through functions on the total space, and this is what the following theorem states.

Definition 7.3.1 (Sections of the associated bundle)**.** *A* section *of E is a map $s : G/H \to E$ with $s(p) \in E_p$ for all $p \in G/H$, which is Borel with respect to the local trivializations of Lemma 7.2.4.*

Theorem 7.1 (Equivariant functions are sections)**.** *The sections of E are in bijection with the Borel functions $f : G \to V$ satisfying the equivariance condition*

$$f(gh) = \pi(h^{-1}) f(g), \qquad g \in G,\ h \in H. \tag{7.5}$$

The bijection assigns to f the section $s(gH) := [g, f(g)]$, and to s the function $f(g) := v$, where $v \in V$ is the unique element with $s(gH) = [g, v]$.

Proof. Let s be a section and let f be the function defined by the second assignment, which is well defined by Lemma 7.2.4. Since $s(gH) = s(ghH)$, the equivalence relation yields

$$[g, f(g)] = s(gH) = s(ghH) = [gh, f(gh)] = [g, \pi(h) f(gh)].$$

By the injectivity of $v \mapsto [g, v]$ it descends that $f(g) = \pi(h) f(gh)$, *i.e.*, f satisfies Equation (7.5).

Conversely, let f satisfy Equation (7.5). The class $[g, f(g)]$ depends only on gH, because for $h \in H$

$$[gh, f(gh)] = [gh, \pi(h^{-1}) f(g)] = [g, \pi(h)\pi(h^{-1}) f(g)] = [g, f(g)],$$

and the formula $s(gH) := [g, f(g)]$ defines a section. The two assignments are inverse to each other by construction. □

Remark 7.3.2. The equivariance condition (7.5) is therefore not an ad hoc requirement. It is exactly the condition under which f defines a section of the associated bundle. The inverse $\pi(h^{-1})$ in place of $\pi(h)$ comes from the equivalence relation (7.4). The space $\mathcal{E}_\pi$ of Definition 6.4.5 is the space of sections of $SL(2, \mathbb{C}) \times_{SU(2)} V$.

7.4 The Action of G on Sections

The group G acts on itself by left translation, and this action commutes with the right action of H used in the quotient. It descends therefore to an action on the associated bundle and, through it, on the sections.

Definition 7.4.1 (Left translation on sections)**.** *For $g \in G$ and a section s of E we set*

$$(g \cdot s)(p) := g \cdot s(g^{-1} p), \qquad p \in G/H,$$

where $g \cdot [g', v] := [gg', v]$ is the action on E induced by left translation on G, and G acts on G/H by left translation of cosets.

The action on E is well defined because $(gg'h, v) \sim (gg', \pi(h)v)$ whenever $(g'h, v) \sim (g', \pi(h)v)$, and it covers the action on the base. The next proposition expresses the action in terms of functions on G, where it takes the form used in the chapter.

Proposition 7.4.2 (The action on equivariant functions)**.** *Under the bijection of Theorem 7.1, the action of G on sections corresponds to the left translation*

$$(\mathcal{L}(g) f)(g') := f(g^{-1} g'), \qquad g, g' \in G. \tag{7.6}$$

In particular $\mathcal{L}(g)$ preserves $\mathcal{E}_\pi$ and $g \mapsto \mathcal{L}(g)$ is a representation of G on $\mathcal{E}_\pi$.

Proof. Let f be the function corresponding to s, so that $s(g'H) = [g', f(g')]$ for every $g' \in G$. Then

$$(g \cdot s)(g'H) = g \cdot s(g^{-1} g' H) = g \cdot [g^{-1} g', f(g^{-1} g')] = [g', f(g^{-1} g')],$$

so that the function corresponding to $g \cdot s$ is $g' \mapsto f(g^{-1} g') = (\mathcal{L}(g) f)(g')$. Since $g \cdot s$ is a section, this function satisfies Equation (7.5), which proves that $\mathcal{L}(g)$ preserves $\mathcal{E}_\pi$. Moreover

$$(\mathcal{L}(g)\mathcal{L}(\tilde{g}) f)(g') = (\mathcal{L}(\tilde{g}) f)(g^{-1} g') = f(\tilde{g}^{-1} g^{-1} g') = (\mathcal{L}(g\tilde{g}) f)(g').$$ □

Remark 7.4.3. The action of G on $\mathcal{E}_\pi$ is by left translation, and the equivariance condition is a constraint on the right action of H. The two actions commute, and this is the structural reason for the canonicity of the construction. No choice of a section of ϖ and no trivialization has entered so far.

7.5 Trivializations and the Role of a Section

The formulation in terms of functions on G is canonical but cumbersome, because G is larger than the base G/H. To pass to functions on G/H one needs a choice of a representative in each coset, *i.e.*, a section of the principal bundle.

Definition 7.5.1 (Section of the principal bundle)**.** *A* section *of $\varpi : G \to G/H$ is a smooth map $s_0 : G/H \to G$ with $\varpi(s_0(p)) = p$ for all $p \in G/H$.*

In the setting of the chapter the standard choice is $s_0(p) = A_p$, the standard boost of Proposition 6.4.1, which is smooth. A global section trivializes the principal bundle, and consequently the associated one.

Lemma 7.5.2 (Trivialization induced by a section)**.** *Let s_0 be a section of ϖ. The map*

$$\Theta_{s_0} : G/H \times H \to G, \qquad \Theta_{s_0}(p,u) := s_0(p)\,u, \tag{7.7}$$

is a diffeomorphism, so that every $g \in G$ is uniquely $g = s_0(p)u$ with $p = \varpi(g)$ and $u \in H$. Moreover $(p,v) \mapsto [s_0(p), v]$ is an isomorphism of vector bundles from $G/H \times V$ onto E.

Proof. The map Θ_{s_0} is smooth. Since $s_0(\varpi(g))H = gH$, one has $s_0(\varpi(g))^{-1} g \in H$ for every g, and $g \mapsto (\varpi(g), s_0(\varpi(g))^{-1} g)$ is a smooth map which is inverse to Θ_{s_0}. For the second claim, $(p,v) \mapsto [s_0(p), v]$ is smooth and linear on the fibres, and it is bijective by Lemma 7.2.4. □

The trivialization allows one to exchange an equivariant function $f : G \to V$ for a function on the base. We set

$$\psi(p) := f(s_0(p)), \qquad f(s_0(p)u) = \pi(u^{-1})\,\psi(p), \tag{7.8}$$

where the second equation is the inverse assignment, and it follows from Equation (7.5). In terms of sections, $s(p) = [s_0(p), \psi(p)]$. For $s_0(p) = A_p$ this is the bijection of Lemma 6.4.6.

Remark 7.5.3. The identification (7.8) depends on the choice of s_0. If s_0' is another section, then $s_0'(p) = s_0(p)h(p)$ with $h(p) := s_0(p)^{-1} s_0'(p) \in H$, and

$$\psi'(p) := f(s_0'(p)) = f(s_0(p)h(p)) = \pi\big(h(p)^{-1}\big)\psi(p).$$

Thus the functions on G/H change by a p-dependent transformation. The description on G is canonical, whereas the description on G/H requires a choice.

7.6 The Transition Function and the Wigner Rotation

We now compute the action of G on the functions ψ obtained from a trivialization. This is where the Wigner rotation appears. The computation needs a quantity measuring how far a section is from being preserved by the left action of G.

Definition 7.6.1 (Transition function)**.** *Let s_0 be a section of ϖ. For $g \in G$ and $p \in G/H$ the* transition function *is the unique element $h(g,p) \in H$ such that*

$$g^{-1} s_0(p) = s_0(g^{-1}p)\,h(g,p). \tag{7.9}$$

The element exists and is unique because $g^{-1}s_0(p)$ lies over $g^{-1}p$, hence in the coset $s_0(g^{-1}p)H$ on which H acts simply transitively. It satisfies a cocycle identity, which is the statement that the transformation law of the next proposition composes correctly.

Lemma 7.6.2 (Cocycle identity)**.** *For $g, g' \in G$ and $p \in G/H$, $h(gg', p) = h(g', g^{-1}p)\,h(g,p)$ and $h(e,p) = e$.*

Proof. By Equation (7.9) applied twice,

$$(gg')^{-1} s_0(p) = g'^{-1} s_0(g^{-1}p)\,h(g,p) = s_0\big(g'^{-1}g^{-1}p\big)\,h(g', g^{-1}p)\,h(g,p),$$

and the uniqueness in Definition 7.6.1 yields the claim. The second identity is immediate. □

Proposition 7.6.3 (The action in a trivialization)**.** *Let $f \in \mathcal{E}_\pi$ and $\psi(p) := f(s_0(p))$. Then the function associated to $\mathcal{L}(g)f$ is*

$$(g \cdot \psi)(p) = \pi\big(h(g,p)^{-1}\big)\,\psi(g^{-1}p). \tag{7.10}$$

Proof. By definition and by Equations (7.9) and (7.5),

$$(g \cdot \psi)(p) = (\mathcal{L}(g)f)(s_0(p)) = f(g^{-1}s_0(p)) = f\big(s_0(g^{-1}p)h(g,p)\big) = \pi\big(h(g,p)^{-1}\big)f\big(s_0(g^{-1}p)\big),$$

which is Equation (7.10). □

Equation (7.10) has the form of the transformation law (6.17) of the chapter. The next proposition identifies the two.

Proposition 7.6.4 (The Wigner rotation is the transition function)**.** *For $G = SL(2,\mathbb{C})$, $H = SU(2)$ and $s_0(p) = A_p$, the transition function is*

$$h(g,p) = R_W(g,p)^{-1}, \tag{7.11}$$

where $R_W(g,p) = A_p^{-1} g A_{\Lambda(g)^{-1}p}$ is the Wigner rotation of Equation (6.16)*. In particular Equation* (7.10) *is Equation* (6.17)*, and Lemma 7.6.2 is the cocycle relation of Proposition 6.4.7.*

Proof. Put $q := \Lambda(g)^{-1}p$. By Definition 7.6.1, $g^{-1}A_p = A_q\,h(g,p)$, hence

$$h(g,p) = A_q^{-1}g^{-1}A_p = \big(A_p^{-1}gA_q\big)^{-1} = R_W(g,p)^{-1}.$$

The cocycle identity of Lemma 7.6.2, written for $R_W = h^{-1}$, reads $R_W(gg',p) = R_W(g,p)R_W(g',\Lambda(g)^{-1}p)$. □

The Wigner rotation has therefore a geometric meaning, which clarifies why it appears and when it can be avoided.

Remark 7.6.5. The transition function measures the failure of the trivialization to be compatible with the left action of G. If $h(g,p)$ does not depend on p, then Equation (7.10) is a fixed matrix acting on $\psi(g^{-1}p)$, which is a local transformation law. This is the case in the Galilean setting. For $G = K \rtimes H$, with K normal, and $s_0(p) \in K$ the representative of p in K, a direct computation yields $h(g,p) = u^{-1}$ for $g = ku$, so that only the rotation part of g enters, and boosts act on $\mathbb{C}^{2j+1}$ trivially. For $G = SL(2,\mathbb{C})$ and $s_0(p) = A_p$, Proposition 6.5.1 shows that $h(g,p) = R_W(g,p)^{-1}$ does depend on p for $g = A_q$, because the product of two boosts with different directions contains a rotation.

7.7 Dictionary and Concrete Examples

The correspondence between the language of induced representations and that of fibre bundles is summarized in Table 7.1. We record it explicitly because it is the key to reading the construction of Section 6.4 geometrically.

The simplest example in which the associated bundle is not trivial shows that the construction contains a familiar piece of mathematical physics.

Example 7.7.1 (Induction from $U(1)$ to $SU(2)$). Take $G = SU(2)$, $H = U(1)$ the subgroup of the matrices $\mathrm{diag}(e^{i\theta}, e^{-i\theta})$, and let $\pi_n(e^{i\theta}) := e^{in\theta}$ be the character labelled by $n \in \mathbb{Z}$. The principal bundle is the Hopf fibration

$$U(1) \hookrightarrow SU(2) \twoheadrightarrow S^2,$$

where $S^2 \cong SU(2)/U(1)$ is the Bloch sphere. The associated bundle $E_n := SU(2) \times_{U(1)} \mathbb{C}$ is a complex line bundle over S^2 with first Chern number $\pm n$, so that it is trivial for $n = 0$ and nontrivial otherwise. By Frobenius reciprocity [**TODO-BarutRaczka**], the space $L^2(S^2, E_n)$ with the action of $\mathcal{L}$ decomposes as the direct sum of the irreducible representations d_j of $SU(2)$ with $j \geq |n|/2$ and $j - n/2 \in \mathbb{Z}$, each of them once. For $n = 0$ the sections are the functions on S^2 and the decomposition is the one into spherical harmonics. For $n \neq 0$ the sections are the monopole harmonics.

Induced representation	**Fibre bundle**
G	Total space of the principal bundle
H	Structure group
G/H	Base of the bundle
(π, V)	Fibre of the associated bundle with its H-action
$f : G \to V$ with $f(gh) = \pi(h^{-1})f(g)$	Section of the associated bundle
Left translation $\mathcal{L}(g)$	Action of G on sections
Section $s_0 : G/H \to G$	Trivialization of the bundle
$h(g,p)$ with $g^{-1}s_0(p) = s_0(g^{-1}p)h(g,p)$	Transition function
$R_W(g,p)^{-1}$	Transition function for $s_0(p) = A_p$
Change of section $s_0 \mapsto s_0 h$	Change of gauge
$\psi(p) = f(s_0(p))$	Section in the trivialization

Table 7.1: Dictionary between induced representations and fibre bundles.

Example 7.7.2 (The Dirac case). Take $G = SL(2,\mathbb{C})$, $H = SU(2)$ and $(\pi, V) = (d_j, \mathbb{C}^{2j+1})$. The principal bundle is the one of Equation (7.2), the associated bundle is $SL(2,\mathbb{C}) \times_{SU(2)} \mathbb{C}^{2j+1}$, and its sections are the elements of $\mathcal{E}_\pi$ of Definition 6.4.5. The left translation $\mathcal{L}$ is the representation of Proposition 6.4.7, and the Wigner rotation is the transition function of the trivialization $p \mapsto A_p$. The representation of the full Poincaré group is obtained by letting the translations act on the sections by multiplication with $e^{ia\cdot p/\hbar}$, as in Definition 6.4.8, and by taking the Hilbert space of square integrable sections for the invariant measure of Lemma 6.4.3.

7.8 The Role of the Construction in the Chapter

We conclude with two remarks on the logical role of the induced representation in the chapter. The first concerns what the construction does, and the second what it leaves open.

Remark 7.8.1 (Canonicity and universality). The construction is canonical, since it requires neither a complement to H in G nor a section of ϖ. The formulation by equivariant functions on G is the canonical one, and the passage to functions on G/H requires a section and introduces the transition function, which is the Wigner rotation. Its universality is the content of the theorem of Wigner and Mackey, Theorem 6.4. Every elementary relativistic system of mass $m > 0$ is unitarily equivalent to exactly one $U_{m,j}$.

Remark 7.8.2 (The problem left open). The representation $U_{m,j}$ is unitary and irreducible, and it contains all the content of an elementary relativistic system. Its defect is that the transformation law (6.17) involves the matrix $\pi(R_W(g,p))$, which depends on the momentum once the state is written in position space. In the language of bundles, the transition function of the standard trivialization is not independent of p. A local transformation law would require a fixed matrix, *i.e.*, a unitary finite-dimensional representation of $SL(2,\mathbb{C})$ acting on the fibre, and Propositions 6.5.9 and 6.5.10 show that none exists. The way out, carried out in Sections 6.5 and 6.6, is to embed the associated bundle E into a trivial bundle $\mathcal{O}_m^+ \times W$, on which $SL(2,\mathbb{C})$ acts through a non-unitary representation D with a fixed matrix. The image is a subbundle with fibres $F(p) \subseteq W$, which depend on p. The subbundle is cut out by the Dirac equation, and its dependence on p is the price paid for locality.